Highlights

The Online Paradox: Why Software Practitioners Feel Less Safe Learning in Online Settings

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- Psychological safety is significantly lower in online versus face-to-face professional learning interactions among software practitioners (Hedges' $g = 1.07$, large effect).
- 63% of practitioners report exclusionary behaviors; practitioners respond by reducing contributions (58%) and self-censoring (20%).
- Gender, age, role, seniority and content creatorship did not significantly influence perceptions of psychological safety in face-to-face and online modalities. However, women did report more negative interactions. Different age cohorts also reported different levels of exclusionary behaviors, hostility and negativity.
- Member checking with 30 practitioners validated findings and revealed that psychological safety enables rather than constrains constructive disagreement.
- Mixed-methods design with 143 participants provides empirical characterization of how interaction modalities shape learning behaviors in (distributed) software engineering work.

The Online Paradox: Why Software Practitioners Feel Less Safe Learning in Online Settings

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Abstract

As software engineering work shifts toward distributed and hybrid models, professional learning increasingly occurs through online, peer-based platforms alongside traditional face-to-face settings. While digital environments expand access to peers and expertise, their impact on the psychological conditions that support learning remains unclear. Psychological safety — the shared belief that one can speak up, ask questions, and disagree without fear of embarrassment or exclusion — is central to learning and innovation, yet underexplored across interaction modalities in software engineering.

This mixed-methods study examines how 143 software practitioners experience psychological safety in online and face-to-face learning interactions. We use a mix of quantitative and qualitative data to investigate how psychological safety varies between modalities and demographic variables, and what contributes to low psychological safety in the experience of practitioners Results show that self-reported psychological safety is substantially lower in online than in face-to-face interactions. Practitioners describe exclusionary behaviors, negative interaction patterns, and hostility as primary threats, and typically respond by reducing contributions, avoiding interpersonal risks, and withdrawing altogether. Member checking was used to validate interpretations, and involve practitioners directly in the sense-making of results,

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surfacing explanatory mechanisms and implications grounded in professional experience.

These findings suggest an online paradox: environments designed to broaden participation may simultaneously make interpersonal risk-taking feel less safe, undermining their learning potential. The study offers an evidence-based characterization of how interaction modalities shape learning-related behaviors in contemporary software work. Psychological safety is positioned as a condition that enables, rather than suppresses, constructive disagreement, and the results inform practitioners and organizers designing hybrid and online learning settings in software engineering.

Keywords: Psychological Safety, Software Engineering, Online Learning, Distributed Work, Professional Development

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1. Introduction

The ongoing transformation of work toward hybrid, AI-supported, distributed, and asynchronous modes is reshaping how knowledge is created, shared, and sustained. Software engineering (SE) in particular is marked by rapid technological and methodological change, which makes the capacity for continuous learning foundational for professional practice [93]. As formal curricula struggle to keep pace, software practitioners increasingly rely on peer-based learning and knowledge exchange within project teams, professional events, and online platforms [88, 87]. The shift to hybrid and distributed work implies that such learning now occurs across both remote, online environments and on-site, face-to-face settings, often extending well beyond organizational boundaries [93]. Empirical studies of software professionals during the COVID-19 pandemic indicate that remote and hybrid arrangements affect well-being, productivity, and collaboration patterns [75, 74], which reinforces the need to understand how learning conditions such as psychological safety operate in these settings.

While the importance of continuous learning in distributed and online environments continues to grow, the conditions that enable effective learning in such modalities are not well understood. One critical enabler for learning already identified is psychological safety [98]. It is commonly defined as a shared belief that one can speak up, ask questions, and express disagreement without fear of embarrassment or exclusion, and it has been linked to learning, collaboration, and innovation [21, 22]. Psychological safety is a condition that enables constructive disagreement, not an absence of conflict. Yet the dynamics that shape psychological safety in loosely structured, voluntary, and often public online settings remain largely unexplored. Participants in such contexts are frequently unknown to one another, operate within flat or ambiguous hierarchies, and may be subject to wide audience exposure, all of which can heighten perceived interpersonal risk. Moreover, when learning interactions are mediated by asynchronous messaging, video calls, or semi-anonymous forums, traditional cues for trust, inclusion, and repair are weakened. As a result, psychological safety may be diminished, leading participants to disengage or withdraw. Paradoxically, the very environments designed to expand access and participation may fail to provide the conditions necessary for effective learning. This paradox motivates a central question for contemporary software practice: *How does psychological safety manifest in online and face-to-face learning interactions, and what does this mean for how practitioners participate?*

This study is motivated by three gaps in existing knowledge:

1. A lack of empirical comparison between psychological safety in online and face-to-face professional learning interactions.
2. Limited understanding of how psychological safety varies across demographic or professional categories such as age, gender, role, or seniority [22].
3. Uncertainty about how practitioners perceive the feasibility and usefulness of strategies intended to improve psychological safety in large, distributed learning settings [66].

To address these gaps, we conducted a two-phase mixed-methods study. In Phase I, we surveyed members of agile software engineering learning settings ($N = 143$), combining quantitative measures of psychological safety across interaction modalities with qualitative accounts of concrete experiences. In Phase II, we carried out member checking with practitioners ($N = 30$) to validate interpretations and refine our understanding of the patterns observed.

The following questions guide our research:

- RQ_1: *How do software practitioners experience psychological safety in online and face-to-face learning interactions?*
- RQ_2: *How does perceived psychological safety relate to practitioners' learning-related behaviors across these modalities?*

Our findings indicate that self-reported psychological safety is substantially lower in online learning interactions than in face-to-face modalities. No significant differences were found between demographic variables of gender, age, seniority, content creatorship and role ($N = 143$). Practitioners describe exclusionary behaviors, unconstructive interaction patterns, and hostility as recurring behavioral threats in online spaces, and respond by reducing contributions, avoiding interpersonal risks, or withdrawing from discussions. At the same time, they emphasize that disagreement and critical debate are essential for learning, and that psychological safety concerns the interpersonal costs of such disagreement rather than its presence or absence. By performing member-checking ($N = 30$), we established that the results are broadly recognized by participants. They actively engaged in interpreting potential causes and consequences. Taken together, the results highlight an "online paradox": environments intended to broaden participation may also make the interpersonal risk-taking required for learning feel less safe.

The remainder of the paper is structured as follows. Section 2 reviews prior work on psychological safety and learning in software engineering and related domains. Section 3 details the research design and mixed-methods approach. Sections 4 and 5 present the quantitative and qualitative findings from the survey and member checking in Phase I and II, respectively. Section 6 discusses implications for software practitioners and organizers of professional learning settings, as well as limitations. Section 7 outlines directions for future research.

2. Related work

Continuous learning has long been recognized as a prerequisite for maintaining competence in software engineering, where technologies, tools, and practices evolve rapidly [93]. Practitioners rely on a mix of formal structures (teams, projects, training) and informal structures (user groups, online forums, conferences, and social media) to acquire and share knowledge [93]. The rise of distributed and hybrid work has intensified this reliance on digitally mediated interaction, as learning now spans co-located, hybrid, and fully online settings, often beyond organizational boundaries. This study contributes to this line of work by examining how psychological safety, a key social condition for learning, is experienced across these interaction modalities.

2.1. Psychological safety in software engineering

Psychological safety is defined as a shared belief that a context is safe for interpersonal risk-taking, such as asking questions, admitting mistakes, or expressing dissenting views [21, 22]. In software engineering, psychological safety has been linked to team effectiveness, learning, and software quality. Lenberg et al. [53] found that psychological safety and team norm clarity both predicted self-assessed team performance and job satisfaction in development teams, with norm clarity emerging as a particularly strong predictor. In a mixed-method study of 6.000 members from 2.000 agile teams, Verwijs & Russo [100] found that psychological safety correlated strongly with other indicators of continuous learning, and contributed indirectly to team effectiveness.

Recent empirical work in agile settings has identified antecedents of psychological safety such as team autonomy, leadership behaviors, and work design characteristics, and has shown that psychological safety is associated with better coordination and higher perceived software quality [7, 98, 101]. High psychological safety supports learning behaviors such as seeking feedback, admitting mistakes, and asking for help [21], and is consistently linked to team effectiveness in software engineering and related domains [62, 94, 42, 100].

This body of work, however, primarily focuses on relatively bounded entities such as project teams or organizational units, where membership, roles, and responsibilities are clear. Much less is known about psychological safety in more open and loosely structured professional learning settings, which include public or semi-public online platforms (e.g., Q&A sites, social media discussions, open communities) alongside conferences and meetups. For example, studies of psychological safety in online developer communities such as Stack Exchange show that perceived safety and norm clarity shape willingness to ask and answer questions, but systematic comparisons across modalities (online versus face-to-face) remain rare [78]. The present study addresses this gap by comparing practitioners' experiences of psychological safety across different modalities rather than within a single team or organization.

2.2. Psychological safety in online and hybrid interaction

Outside software engineering, research on online and hybrid teams suggests that psychological safety is more fragile when interaction is mediated by digital tools. Non-verbal cues, opportunities for informal repair, and shared situational awareness are reduced, which can increase the perceived interpersonal cost of speaking up [50, 41]. Conceptual and empirical work on virtual teams highlights trust, explicit communication norms, and deliberate relationship-building as key enablers of psychological safety in remote settings [52]. Case-study work on virtual teams further underlines that psychological safety becomes more salient as physical co-presence is reduced.

In software engineering, empirical accounts of remote and hybrid work have documented similar challenges, such as difficulties in establishing trust, uneven participation in online meetings, and the prevalence of public criticism in open channels [93]. While some studies have begun to examine psychological safety in distributed software teams [98], systematic, peer-reviewed evidence about how psychological safety is experienced across modalities in software practitioners' learning interactions remains limited. This study complements existing team-focused work by examining psychological safety across the broader ecology of professional learning settings that practitioners inhabit.

2.3. Continuous learning, participation, and interpersonal risk

Continuous learning in software engineering is often enacted through participation in professional communities, both local (e.g., meetups, user groups) and global (e.g., online forums, social media, large-scale conferences) [93]. Such contexts offer access to diverse perspectives, emerging practices, and problem-specific advice, but they also expose practitioners to public scrutiny and potential status loss. Prior work in organizational learning has emphasized that psychological safety underpins learning behaviors such as asking for help, seeking feedback, and voicing concerns [21, 46]. In software engineering, similar behaviors are essential for practices such as code review, pair programming, and collaborative problem solving.

At the same time, prior research stresses that psychological safety does *not* imply the absence of disagreement or conflict. Rather, it concerns whether disagreement can be expressed without fear of humiliation or exclusion [23]. From this perspective, constructive conflict and critical debate are compatible with, and even indicative of, psychologically safe environments [23, 101]. This nuance is particularly important in professional learning settings, where strong opinions and critical feedback are common, especially in online channels.

Existing work has begun to connect psychological safety to specific software practices. For instance, studies suggest that team autonomy and clarity of norms influence psychological safety in agile teams [7], and that psychological safety mediates relationships between leadership behaviors and innovation outcomes. However, most of these studies do not explicitly examine how the *interaction modality* itself (e.g., face-to-face versus online, synchronous versus asynchronous) shapes practitioners' perceptions of psychological safety and their participation choices. Our study addresses this gap by providing an empirical comparison of psychological safety across online and face-to-face learning interactions for software practitioners, and by characterizing how perceived safety relates to behaviors such as contributing, asking questions, and withdrawing.

2.4. Research gap and hypotheses

Taken together, prior work suggests that psychological safety is an important condition for learning and collaboration in software engineering teams, and that it can be challenging to sustain in online and hybrid settings [21, 23, 53, 7, 98]. However, existing studies rarely compare how the *same* practitioner population experiences psychological safety across online and face-to-face learning interactions, nor do they systematically examine how demographic and professional characteristics shape these perceptions in such contexts. Building on this literature, we formulate two hypotheses.

First, research on virtual teams and remote software work indicates that reduced non-verbal cues, public visibility, and asynchronous interaction can increase the perceived interpersonal cost of speaking up [50, 41, 52, 98]. We therefore expect psychological safety to be lower in online learning interactions than in face-to-face ones:

H1. *Psychological safety is lower in online learning interactions than in face-to-face learning interactions for software practitioners.*

Second, prior work on psychological safety highlights demographic and role-related moderators, but empirical findings are mixed and often based on organizational teams rather than broader professional learning settings [22, 7, 53]. We do not expect large or systematic differences across demographic or professional categories:

H2. *Psychological safety in learning interactions does not differ substantially across (a) gender, (b) age group, (c) content creatorship, (d) professional role, and (e) seniority in the professional community.*

3. Research Design

This study aims to characterize how software practitioners experience psychological safety in professional learning interactions across different modalities. It adopts a two-phase mixed-methods design comprising a survey study (Phase I) and a validation and refinement phase based on member checking (Phase II).

Phase I consisted of an online survey targeting software practitioners who regularly participate in professional learning settings, such as meetups, conferences, online communities, and discussion platforms. The survey combined quantitative measures of psychological safety for online and face-to-face interactions with qualitative questions about concrete experiences of low psychological safety, its perceived effects on participants, contributing factors, and perceived improvement strategies. The use of both quantitative and qualitative components follows established mixed-methods traditions [16]. Quantitative data enable hypothesis testing and estimation of the magnitude of differences across modalities and groups [36], while qualitative data provide richer insight into how practitioners interpret and enact psychological safety in their everyday learning interactions [17]. Bringing these strands together supports methodological triangulation and can strengthen the credibility and robustness of inferences by converging evidence from complementary sources [37]. The design and implementation of Phase I are described in detail in Section 4.

Because psychological safety is a subjective and context-dependent experience, Phase II focused on validating and refining the interpretations from Phase I through *member checking*. Lincoln and Guba describe member checking as the process of sharing preliminary findings with participants to solicit feedback, assess credibility, and identify misinterpretations [54]. Subsequent work emphasizes member checking as a way to integrate participants' expertise into the analysis and to surface nuances that may be overlooked by researchers alone [59, 97]. In this study, a subset of practitioners was invited to review key quantitative and qualitative results, reflect on whether these patterns resonated with their experiences, and suggest refinements or alternative interpretations. Given the professional learning focus, this phase also provided an opportunity to co-interpret the implications for practice. The design and execution of Phase II are detailed in Section 5.

The main steps in the overall design are summarized in Figure 1. A replication package is available on Zenodo to support transparency and enable secondary analyses and extensions of this work.

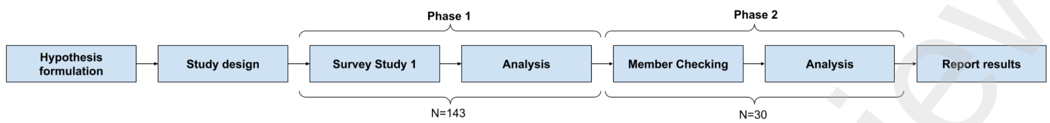

Figure 1: Overview of the two-phase mixed-methods design.

4. Phase I: Survey study

4.1. Participants

Data for Phase I were collected through an online survey deployed using IBM Qualtrics between December 2023 and March 2024. We used a purposive non-probabilistic sampling strategy [3] aimed at recruiting software practitioners who regularly participate in Agile-related professional learning settings. Recruitment channels included Agile-focused communities on LinkedIn, online and in-person meetup groups, professional conferences, relevant podcasts, and newsletters. In addition, several well-known community organizers promoted the study within their networks and to participate themselves.

The survey employed Likert-scale items to collect quantitative data and essay-style questions to collect qualitative data. The complete Phase I survey is provided in Table A.22 and Table A.23 in Appendix A. To reduce self-selection bias, particularly among individuals who might already feel unsafe, we did not collect personally identifiable information (e.g., name, email address) and we suppressed subgroup analyses for groups with fewer than three respondents [60]. These protections were explained in the survey introduction and all invitations.

In total, 159 responses were received. Because recruitment was performed anonymously due to the sensitive subject matter, and relied on open community channels, a response rate could not be computed. Following recommended practices for detecting careless or insufficient effort responses [60], we removed 16 responses that were completed in less than 180 seconds or exhibited patterns indicative of low engagement. The final dataset for Phase I consists of 143 participants. The distribution of the sample by gender, age group, content creatorship, role, and seniority is reported in Table 1.

Table 1: Sample Composition, Phase I (N=143)

Variable	Category	N	%
Gender	Women	43	30.1
	Men	90	62.9
	Other/Prefer not to say	10	7.0
Age (years)	26-35	14	9.8
	36-45	44	30.8
	46-55	54	37.8
	56-65	21	14.7
	66+	5	3.5
	Prefer not to say	5	3.5
Role	Scrum Master	35	24.5
	Agile Coach	49	34.3
	Trainer/Facilitator	17	11.9
	Consultant	21	14.7
	Other	18	12.6
	Prefer not to say	3	2.1
Seniority	$<$ 2 years	3	2.1
	2-5 years	14	9.8
	5-10 years	56	39.2
	$>$ 10 years	70	49.0
Content Creator	Yes	87	60.8
	No	56	39.2

4.2. Measurements

Psychological safety: Psychological safety is typically operationalized in the context of a bounded group with stable membership and task interdependence, emphasizing learning behaviors such as admitting mistakes, experimenting with new ideas, and challenging established practices within ongoing work teams [21, 24]. This study examines psychological safety in learning interactions within professional learning settings, which differ from teams in important ways. Learning interactions are often episodic, involve participants without prior relationships, and occur in contexts with fluid or ill-defined boundaries, asymmetric expertise, and limited social cues, particularly online [103, 93]. In such settings, interpersonal risks are less about coordinated task failure and more about social evaluation, legitimacy, and reputational harm.

Our operationalization thus emphasizes participants' perceived ability to ask for help, voice differing views, and have their contributions taken seriously, rather than behaviors related to task failure and coordination. At its core, both operationalizations are concerned with the anticipated interpersonal cost of speaking up [23]. However, what constitutes "speaking up" varies by context. In professional learning communities, it involves exposing knowledge gaps or offering tentative perspectives to (relative) strangers instead of task failure. This aligns with calls to treat psychological safety as a context-sensitive construct, whose behavioral indicators depend on the nature of interaction and participation [67].

We developed a three-item scale to capture learning behaviors associated with psychological safety [21], tailored to learning interactions in professional communities. The scale comprised three items: (1) "Members of the Agile community foster an environment where it is easy to ask for help," (2) "Members of the Agile community are open to different perspectives and opinions," and (3) "People in the Agile community make an effort to understand each other's views.". A version of this scale was piloted between June and December 2023 using three high-loading items adapted from Edmondson's original measure to assess convergent validity. The subscales showed a strong correlation ($r = .81, N = 131$). A principal component analysis (PCA) indicated a single-factor solution, with all item loadings exceeding .70. The scale demonstrated good internal reliability ($\alpha = .84$).

For the present study, two additional items were included. One item assessed the perceived importance of psychological safety ("It is important to me that interactions in the Agile community are psychologically safe"), and a second item was added to assess face validity ("I experience psychological safety in the Agile community to be high").

Participants completed the scale separately for face-to-face and online interaction modalities. For the online modality, participants were instructed to consider the online medium they used most frequently (e.g., LinkedIn, Slack, Discord, or online forums). This improves ecological validity and interpretability by anchoring responses in the online environment most familiar to participants. To reduce order effects, item presentation was randomized. The reliability of the original three-item scale was high for both modalities (face-to-face: $\alpha = .86$; online: $\alpha = .85$).

Qualitative questions: Four essay questions were included to explore the qualitative experience of low psychological safety, its effects on participants, and strategies to improve it.

- "What are examples of behavior you have personally encountered during interactions in the Agile community that lowered psychological safety for you?"
- "How does a lack of psychological safety in interactions in the Agile community affect you or your behavior?"
- "In your experience, what behavior or factors decrease psychological safety in the Agile community?"
- "What measures or strategies could improve psychological safety and encourage more respectful and constructive discussions within the Agile community?"

We did not distinguish between modalities for these questions to avoid overloading participants. We analyzed responses using the qualitative coding approach described in Section 4.3.

Gender: The gender of participants was measured with a single categorical item "What is your gender?". The available options were "man", "woman", "non-binary / third gender" and "prefer not to say". During analyses, non-binary and third genders were merged with "other" as their number ($N = 1$) was too small for meaningful group comparison.

Age group: The age group of participants was measured with a single categorical item ("What is your age group?") that invited participants to self-categorize into 10-year buckets: "18-25 years", "26-35 years", "36-45 years", "46-55 years", "56-65 years", "65+ years" and "prefer not to say".

Role: Participants were asked to self-categorize into the role that best described their primary professional function ("What best describes your role?"). Because the study focused on Agile and software development contexts, the response options were: "developer, analyst, tester, designer", "scrum master", "agile coach", "trainer or facilitator", "consultant", "other", and "prefer not to say". We chose to add "consultants" as a separate role, because they are typically hired as external experts, whereas "scrum masters" and "agile coaches" are typically part of teams or organizations. During analyses, developers were merged with "other" as their number ($N = 4$) was too small for meaningful group comparison.

Seniority: Seniority in the Agile professional community was measured with a single categorical item ("How many years have you been active in the

Agile community?"). The available options were: "less than a year", "between 1 and 2 years", "between 2 and 5 years", "between 5 and 10 years", and "more than 10 years".

Interaction setting for professional communities: Several questions were included to gain insight into where and how participants interact with Agile-related professional communities. First, a set of categorical items asked participants how often they shared something on the following online platforms: "LinkedIn", "Twitter / X", "Medium", "Mastodon", "Slack / Discord", "Facebook", "forums", and "YouTube". The platforms were selected based on the authors' experience as active members of Agile and software communities. Response options were: "at least once per day", "at least once per week", "at least once per month", "less than once a month", and "never".

Participants were also asked how often they interacted with such communities in face-to-face or synchronous group settings. The settings were: "conferences", "online meetups", "meetups", "workshops or training", and "other". "Online meetups" were included here because they involve direct, synchronous interaction that is more similar to face-to-face meetings than to asynchronous or large-scale online discussions. For these settings, the frequency categories were "very often", "often", "rarely", and "never". Interval-specific categories (e.g., per week or per month) were not used, as it is unlikely that practitioners attend conferences, workshops, or meetups with the same regularity as they engage with online platforms.

Interaction Frequency: Participants that interact more frequently may report different levels of psychological safety, and this may bias results. Two control variables were introduced to address this. Both were calculated as the sum of frequencies with which participants reported to be active on online platforms and in face-to-face settings as described in the previous measurement. The standardized value was used in subsequent analyses.

4.3. Analysis of the Qualitative Data

Qualitative data were analyzed following a naturalistic inquiry paradigm using iterative constant comparison [35]. Constant comparison guided how codes and interpretations were continuously compared across cases and refined over successive iterations, rather than constituting a standalone analytic method. The overarching analytic approach was thematic analysis, which was used to identify, refine, and report recurring patterns in the qualitative data and to inductively develop a conceptual model of psychological safety experiences [96, 6]. Thematic analysis provided the substantive analytic lens, while constant comparison supported iterative sensemaking throughout the process.

To reinforce methodological rigor, transparency, and traceability, we employed the Gioia et al. framework [34] as a structuring and reporting scaffold for the thematic analysis. This framework is increasingly used in qualitative research to make inductive coding processes explicit and reproducible [38]. Specifically, the Gioia framework was used to organize the results of the thematic analysis into a data structure consisting of three levels. In the first stage, open coding was employed to identify first-order concepts closely aligned with participants' own language. In the second stage, related first-order concepts were grouped into second-order themes through axial coding. In the third stage, these themes were aggregated into higher-level dimensions that capture core aspects of the psychological safety experience. Consistent with Gioia et al., the first two stages were conducted with minimal reliance on prior theory to reduce confirmation bias [34]. A total of 11,415 words were analyzed and coded. Full examples of the thematic analysis are included in Table A.19 in Appendix A.

To strengthen validity, multiple coders were involved in the coding process. The study's first author developed an initial codebook using QualCoder [18]. The second author independently coded all cases using this codebook, while the corresponding author supervised and assessed the overall analytical process. Codes, themes, and dimensions were iteratively memoed and refined until theoretical saturation was reached [81].

Inter-rater reliability was assessed using Cohen's Kappa at the level of first-order concepts and higher-order groupings. For the four main themes, high agreement was achieved after iterative refinement:

- *Experiences of low psychological safety:* 51 first-order concepts, 4 iterations, $\kappa = .92$
- *Effects of low psychological safety:* 67 first-order concepts, 2 iterations, $\kappa = .94$
- *Factors contributing to low psychological safety:* 67 first-order concepts, 2 iterations, $\kappa = .93$
- *Improvement strategies:* 54 first-order concepts, 2 iterations, $\kappa = .95$

This iterative process resulted in a thematic map characterizing how software practitioners experience psychological safety in professional learning interactions.

4.4. Analysis of the Quantitative Data

The quantitative data was analyzed using IBM SPSS and IBM AMOS [1]. Outlier detection was performed with stem-and-leaf plots. Five outliers were identified for the reported psychological safety in face-to-face interactions, but none were found for online interactions. Since the outliers appeared to be legitimate cases with low safety rather than data errors, we retained them. However, we reran statistical tests without outliers and report differences where relevant.

We checked for missing data and found between two and three cases missing per continuous variable. Little's MCAR test confirmed the data were missing completely at random ($\chi^2 = 27.091$, $df = 28$, $p = 0.53$) [39]. We used listwise deletion where relevant.

To **compare psychological safety between modalities within subjects**, we used a paired samples t-test on participants' psychological safety scores. Although individual Likert items are ordinal, psychological safety was operationalized as a composite scale derived from multiple Likert items. Simulation studies have shown that 5-point Likert scales can be treated as continuous in parametric tests [44]. Before comparing psychological safety between modalities, we assessed if psychological safety was conflated with properties specific to settings. Because our measures asked participants to rate psychological safety in the setting they frequented the most, psychological safety could be biased by properties of the settings (i.e. size, public or private, asynchronous or synchronous) rather than the modality itself. A non-parametric Kruskal-Wallis test was used because groups of settings differed substantially in size and variance was not homogeneous. The test did not reveal significant differences between settings for online ($\chi^2(6) = 8.20$, $N = 143$ $p = .224$) and face-to-face modalities ($\chi^2(5) = 2.04$, $N = 143$ $p = .843$). Consequently, there is no indication of conflation by setting. We then assessed the assumptions of the t-test by examining skewness (< 2), kurtosis (< 3), and Q–Q plots [39] and found no issues. A Kolmogorov-Smirnov test revealed a significant deviation from normality, which could introduce bias in our estimates. To mitigate potential violations of the normality assumption and to obtain more robust standard errors and confidence intervals, we employed bootstrap resampling procedures [26]. An effect size based on Hedges' g with a 90% confidence interval was included to calculate the distance between two distributions of within-subject data [89] and allow an interpretation of the meaningfulness of any deviation [99, 47]. Effect sizes below .01 indicate no effect. Effect sizes of 0.02, 0.06, and 0.14 are considered small, moderate, and large in magnitude, respectively.

Structural Equation Modeling (SEM) was used **to test for effects of the demographic variables gender, age, role, seniority and content creatorship on psychological safety**. Structural Equation Modeling allows researchers to analyze latent factors measured through multiple observed indicators, and explicitly estimates measurement error. This provides more accurate estimates than methods relying on Ordinary Least Squares (OLS) such as ANOVA and linear regression [39]. Moreover, SEM is inherently a confirmatory approach that combines multiple linear regression and confirmatory factor analysis (CFA) with Maximum Likelihood estimation (ML). The statistical model is evaluated in SEM with "Goodness of Fit" indices and the statistical significance and effect size of individual paths. The aim is to arrive at a model as parsimonious as possible while providing a good fit (and thus explanatory power). We discuss the fit indices in section 4.4.1.

Prior to estimating the structural equation model, we evaluated whether the data met the key statistical assumptions required. Normality was assessed by examining skewness and kurtosis for all independent, dependent, and control variables. All values fell within the recommended thresholds for skewness (< 2) and kurtosis (< 3), indicating acceptable univariate normality [19].

Homoscedasticity was evaluated through visual inspection of scatterplots for all combinations of continuous independent and dependent variables. No systematic patterns or heteroscedasticity were observed. Multicollinearity was assessed by entering all continuous independent variables individually into a linear regression model [30]. Variance Inflation Factor (VIF) values remained well below the critical threshold of 10[39], ranging from 1.02 to 1.26, indicating no multicollinearity concerns.

Age group and seniority in the community were measured as ordinal variables. Although treating ordinal variables as continuous can result in more parsimonious models and simpler interpretation, this approach requires a monotonic relationship with the dependent variables [57]. To evaluate this assumption, we conducted one-way ANOVAs with polynomial contrasts. These analyses revealed no significant linear, quadratic, or cubic trends for either age group or seniority with respect to psychological safety in online or face-to-face contexts, thereby violating the monotonicity assumption. Additionally, the category intervals for seniority were unequal. Consequently, treating these variables as continuous would have introduced bias. Both age group and seniority were therefore modeled using dummy variables. To avoid perfect multicollinearity, k–1 dummy variables were included, with the first category serving as the reference group.

Because both independent and dependent variables were measured using a single survey instrument, common method bias was considered [69]. Such bias may arise from factors such as social desirability, recall effects, or respondent mood. To account for shared method variance, a common latent factor (CLF) was incorporated into the model following established procedures [69]. The CLF was specified to load on all relevant indicators. As standardized regression weights differed substantially between models with and without the CLF, the factor was retained in the final model.

We estimated a full latent variable model that integrated both the measurement and structural components. The measurement model specifies the relationships between observed indicators (survey items) and their corresponding first-order latent constructs, effectively constituting a confirmatory factor analysis (CFA) [49]. The structural model represents the hypothesized relationships among latent variables and corresponds to a regression framework. Estimating the full latent model reduces convergence problems associated with low indicator reliability and provides greater analytical flexibility and degrees of freedom compared to non-latent approaches [40].

Model estimation followed established guidelines [8, 76, 39]. The within-subject constructs psychological safety (online) and psychological safety (face-to-face) were specified as latent variables and allowed to covary, with their respective survey items serving as indicators. After confirming adequate fit of the measurement model (see Section 4.4.1), the structural paths were added.

Age group and seniority were entered as sets of dummy variables with direct paths to both psychological safety constructs. Dummy variables were also created for the categorical variables gender, role, and Content creatorship and specified with direct paths to Psychological Safety (online) and Psychological Safety (Face-to-Face). The control variables Interaction Frequency (Online) and Interaction Frequency (face-to-face) were entered as observed variables with direct paths to the corresponding psychological safety indicators to partial out variance attributable to interaction frequency. Covariances were freely estimated among all exogenous variables to account for shared variance unrelated to the focal relationships.

In line with methodological recommendations [8, 49], parameter estimates and significance tests were obtained using a bootstrapping procedure with 2,000 resamples and 95% bias-corrected confidence intervals, providing more robust estimates of standard errors, factor loadings, and direct effects.

We performed a post hoc power analysis using G*Power [27], version 3.1.9. We determined that our sample for Phase II allows us to correctly capture small effects ($d = .020$) with a statistical power of 84% ($1 - \beta = .840$) for the sample of 143 participants, and a power of 99.8% ($1 - \beta = 1.000$) for moderate effects ($d = .050$). Thus, we are confident that we can detect moderate effects and most small effects.

4.4.1. Model fit evaluations

Overall model fit was assessed using multiple indices recommended in the structural equation modeling literature [49, 8, 39]. Specifically, we report the Comparative Fit Index (CFI), Tucker–Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR).

The CFI [5] evaluates model fit relative to a null model while accounting for sample size and is among the most widely used incremental fit indices [39]. Values of .97 or higher indicate good fit [8, 39, 4]. The RMSEA [91] incorporates both sample size and a parsimony correction, favoring more parsimonious models with comparable explanatory power [49]. RMSEA values below .08 indicate acceptable model fit [8, 39]. In line with best practices, we report RMSEA confidence intervals in addition to point estimates [11]. The SRMR reflects the standardized average discrepancy between observed and model-implied covariances [39], with values below .08 indicating good fit. Finally, the TLI is an incremental fit index similar to the CFI that compares the hypothesized model to a baseline model with uncorrelated variables; values of .97 or higher are generally considered indicative of good fit [39]. In addition to global fit indices, we evaluated local fit by inspecting standardized residuals, using a cut-off value of 2.58 to identify areas of poor local fit [8].

The initial measurement model showed inadequate fit to the data ($\chi^2(53) = 139.658$; $TLI = .893$; $CFI = .914$; $RMSEA = .107$; $SRMR = .078$). Results of the Confirmatory Factor Analysis (CFA), reported in Appendix A.21, indicated that all items primarily loaded on their intended factors. However, two pairs of indicators — Psychological Safety (online) 4 and 5 and Psychological Safety (face-to-face) 4 and 5 — were removed because their factor loadings fell below the recommended threshold of .70.

Items included to establish face validity (e.g., "I experience psychological safety in the Agile community to be high") demonstrated strong factor loadings in both modalities ($r = .82$ for online and $r = .87$ for face-to-face) and were therefore retained. The revised measurement model consisted of four indicators for psychological safety (online) and four indicators for psychological safety (Face-to-Face).

We subsequently assessed reliability, convergent validity, and discriminant validity. Discriminant validity was evaluated using the heterotrait–monotrait ratio of correlations (HTMT), computed with a third-party AMOS plugin [31], following established guidelines [39, 43]. HTMT values below .90 indicate adequate discriminant validity; this criterion was met for both psychological safety constructs. Convergent validity was assessed using composite reliability (CR) and average variance extracted (AVE). AVE values exceeded the recommended threshold of .50 for all constructs, and CR values were .70 or higher, indicating adequate reliability [39]. Local fit was further examined by inspecting the standardized residual covariance matrix. Residuals exceeding 2.58 indicate that an item may not adequately measure its intended construct [8]. No such residuals were observed. The final measurement model demonstrated good fit to the data ($\chi^2(19) = 31.534$; $TLI = .978$; $CFI = .985$; $RMSEA = .069$; $SRMR = .037$).

We next evaluated the structural (path) model. The model exhibited good fit ($\chi^2(122) = 140.336$; $TLI = .973$; $CFI = .990$; $RMSEA = .033$; $SRMR = .028$; see Table 2). The predictors accounted for 18.7% of the variance in psychological safety (online) and 18.2% in psychological safety (face-to-face). According to conventions in the social sciences, explained variance between 16% and 26% is considered moderate [12].

Finally, we compared the full model, including control variables for interaction frequency, with a model excluding these controls. A chi-square difference test was not significant ($\chi^2(8) = 15.254, p = 0.054$), indicating that the inclusion of the control variables did not significantly improve model fit.

4.5. Results

4.5.1. Where people interact

This study examines psychological safety in software engineering professional learning settings across different interaction modalities. Table 3 summarizes how often participants reported sharing content in various online environments. LinkedIn emerges as the dominant platform, with 33.9% of participants contributing at least weekly, followed by Slack / Discord (23.2%)

Table 2: Structural Equation Model Fit Indices

Index	Value	Threshold	Assessment
χ^2 (df)	140.336 (122)	–	–
CMIN/df	1.150	<3.0	Excellent
RMSEA	.033	<.08	Excellent
RMSEA 90% CI	[.000, .055]	–	–
CFI	.990	>.97	Excellent
TLI	.973	>.97	Excellent
SRMR	.028	<.08	Excellent
R^2 (Online)	18.7%	–	Moderate
R^2 (F2F)	18.2%	–	Moderate

F2F=Face-to-face. Thresholds from Hair et al. (2019).

and forums (7.9%). These data suggest that, for this sample, professional social media and chat-based communities constitute the primary venues for online learning-related contributions.

Table 3: Frequency by which participants share in online settings (N = 143).

Platform	Never	Less than monthly	Monthly	Weekly	Daily
LinkedIn	10.0%	32.3%	23.8%	23.1%	10.8%
Slack / Discord	47.3%	17.8%	11.6%	17.8%	5.4%
Forums	57.1%	26.4%	8.6%	7.9%	0.0%
Twitter / X	69.8%	18.6%	3.9%	7.0%	0.8%
Facebook	73.0%	15.6%	4.3%	5.7%	1.4%
YouTube	68.8%	19.9%	7.8%	2.8%	0.7%
Medium	74.4%	18.6%	3.9%	2.3%	0.8%
Reddit	85.2%	10.9%	0.8%	2.3%	0.8%
Mastodon	93.0%	2.3%	1.6%	1.6%	1.6%

Table 4 reports the frequency with which participants engage in face-to-face or synchronous group settings. Workshops and training sessions, online meetups, and conferences are the most frequently attended contexts, with

55.6%, 57.1%, and 42.9% of participants, respectively, indicating that they participate at least "often". Together, these patterns indicate that practitioners' learning interactions are distributed across a heterogeneous mix of online and face-to-face formats, with no single modality dominating all learning activity.

Table 4: Frequency by which participants interact in face-to-face or synchronous group settings (N = 143).

Setting	**Never**	**Rarely**	**Often**	**Very often**
Workshops or training	4.2%	40.1%	38.0%	17.6%
Online meetups	10.6%	32.4%	44.4%	12.7%
Conferences	12.0%	45.1%	32.4%	10.6%
Other	21.8%	38.7%	29.0%	10.5%
In-person meetups	7.8%	53.2%	28.4%	10.6%

4.5.2. Group differences in quantitative measures of psychological safety

Overall, psychological safety is higher in face-to-face interactions ($M = 3.965$, $SD = .92$, $N = 141$) than in online interactions ($M = 3.006$, $SD = 1.08$, $N = 143$). The difference was statistically significant ($p < .001$) and supports Hypothesis 1. The overall effect size is qualified as *large*, $Hedges'g = 1.071, 90\%CI[.732, 1.057]$. Table 5 shows the effect of demographic variables for psychological safety in face-to-face and online modalities, controlled by respectively face-to-face interaction frequency and online interaction frequency. Factor loadings are reported in Table 6, and control variable effects in Table 7.

The results show no significant effects of gender (H2a), age (H2b), content creatorship (H2c), role (H2d) and seniority (H2e) on psychological safety on online and face-to-face settings, supporting their associated Hypotheses.

We now turn to the results from our qualitative analyses.

Table 5: Structural Equation Model: Effects of Demographic Variables on Psychological Safety

	Face-to-Face				**Online**			
Parameter	b	95% CI	SE	β	b	95% CI	SE	β
Gender effects (H2a) — Reference: Other/Prefer not to say								
Female	0.134	[-.891, 1.172]	0.310	0.078	0.293	[-.379, 1.049]	0.390	0.136
Male	0.319	[-.661, 1.295]	0.302	0.197	0.416	[-.199, 1.079]	0.379	0.204
Age group effects (H2b) — Reference: 26–35 years								
36–45 years	0.052	[-.422, .524]	0.259	0.031	−0.699	[-1.393, .100]	0.325	−0.328
46–55 years	0.154	[-.292, .588]	0.264	0.095	−0.431	[-1.102, .316]	0.331	−0.213
56–65 years	0.130	[-.484, .591]	0.305	0.059	0.133	[-.645, .942]	0.382	0.048
65+ years	0.287	[-.808, 1.483]	0.431	0.067	0.402	[-.819, 1.717]	0.534	0.075
Prefer not to say	−1.000	[-1.984, .513]	0.475	−0.234	−1.146	[-2.192, .085]	0.593	−0.214
Content creatorship (H2c) — Reference: No								
Creator (Yes)	−0.027	[-.333, .253]	0.148	−0.017	−0.060	[-.548, .344]	0.195	−0.030
Role effects (H2d) — Reference: Other								
Developer	−0.261	[-1.437, .593]	0.458	−0.055	−0.193	[-1.435, 1.986]	0.573	−0.032
Scrum Master	0.209	[-.358, .781]	0.248	0.115	0.388	[-.387, 1.170]	0.311	0.170
Agile Coach	−0.020	[-.581, .548]	0.235	−0.012	0.026	[-.749, .803]	0.294	0.012
Trainer/Facilitator	−0.224	[-.944, .451]	0.285	−0.092	0.024	[-.879, .862]	0.364	0.008
Consultant	0.260	[-.294, .834]	0.273	0.117	0.120	[-.699, .917]	0.341	0.043
Seniority effects (H2e) — Reference: < 2 years								
2–5 years	−0.651	[-1.308, .175]	0.510	−0.246	−0.096	[-1.099, .711]	0.638	−0.029
5–10 years	−0.516	[-1.144, .264]	0.480	−0.321	−0.035	[-.929, .636]	0.600	−0.017
10+ years	−0.631	[-1.320, .101]	0.498	−0.402	−0.102	[-.961, .633]	0.621	−0.052

Note. $N = 143$. b = unstandardized coefficient; β = standardized coefficient; SE = standard error; CI = confidence interval. Reference categories shown for each categorical predictor.
No effects reached statistical significance at $\alpha = .05$.

Table 6: Measurement Model: Factor Loadings for Psychological Safety Scales

Indicator	b	95% CI	SE	λ
Psychological safety (face-to-face)				
Item 1: Open to different perspectives	1.000*	[.614, .861]	—	0.776
Item 2: Foster helpful environment	1.000*	[.705, .885]	—	0.820
Item 3: Value unique skills	1.176*	[.794, .961]	0.090	0.904
Item 6: High psychological safety	1.000*	[.490, .846]	—	0.710
Psychological safety (online)				
Item 1: Open to different perspectives	0.958*	[.620, .821]	0.093	0.733
Item 2: Foster helpful environment	1.000*	[.720, .894]	—	0.822
Item 3: Value unique skills	0.958*	[.719, .893]	0.079	0.828
Item 6: High psychological safety	1.000*	[.735, .892]	—	0.834

Note. $N = 143$. b = unstandardized loading; λ = standardized loading; SE = standard error; CI = confidence interval for standardized loading.
Items 4 and 5 were removed due to factor loadings below .70.
$^*p < .001$.

4.5.3. Which behaviors decrease psychological safety?

To better understand how psychological safety is experienced qualitatively, participants were asked to describe behaviors that lowered psychological safety in their interactions. As described in Section 4.3, responses were analyzed with the Gioia methodology [33], yielding a structure of first-order concepts, second-order themes, and third-order aggregate dimensions. The resulting schema, including the percentage of cases associated with each code, is shown in Figure A.4 in Appendix A.

Four aggregate dimensions emerged. The most frequently reported was *exclusionary behaviors* (63.1%), comprising the themes *dogmatism* (43.8%) and *dismissiveness* (35.4%). The second most frequent dimension was *negative interaction patterns* (33.8%), which includes *unconstructive criticism* (18.5%), *posturing* (9.2%), *impoliteness* (7.7%), *discrimination and sexism* (5.4%), and *insensitivity* (4.6%). A further 33.1% of participants reported codes in the dimension *hostile behavior*, including *personal attacks* (20.8%) and other forms of *aggression* (14.6%). The fourth dimension, *unsafe environment* (8.5%), captured behaviors such as *gossiping* and a generally poor discussion culture (5.4%), as well as *negative use of humor* (1.5%) and *un-*

Table 7: Control Variable Effects: Interaction Frequency on Psychological Safety Indicators

Path	b	95% CI	SE	β
Face-to-Face Interaction Frequency				
→ PS F2F Item 1	0.081	[-.091, .263]	0.090	0.080
→ PS F2F Item 2	1.000[a]	[-.128, .211]	—	0.031
→ PS F2F Item 3	−0.015	[-.160, .184]	0.090	−0.015
→ PS F2F Item 6	1.000[a]	[-.145, .240]	—	0.033
Online Interaction Frequency				
→ PS Online Item 1	0.184	[-.054, .318]	0.119	0.142
→ PS Online Item 2	1.000[a]	[-.128, .211]	—	0.031
→ PS Online Item 3	−0.015	[-.160, .184]	0.090	−0.015
→ PS Online Item 6	1.000[a]	[-.145, .240]	—	0.033

Note. $N = 143$. b = unstandardized coefficient; β = standardized coefficient; SE = standard error. Control variables included to partial out variance explained by interaction frequency.

[a]Parameter not estimated separately in the model.

ethical behavior (1.5%). Finally, 9.2% of participants reported no experience with behaviors that lowered psychological safety.

In summary, participants predominantly experience lowered psychological safety when contributions or contributors are explicitly or implicitly devalued. Exclusionary and hostile behaviors—rather than mere disagreement—are central: psychological safety diminishes when ideas are dismissed, people are attacked, or interaction patterns signal that some voices are unwelcome.

4.5.4. How low psychological safety affects participants

Participants also described how low psychological safety affected their own behavior and experience. The Gioia schema for these effects, with associated frequencies, is shown in Figure A.3 in Appendix A.

Six aggregate dimensions emerged from the coding. The most frequently reported was *reduce contributions* (58.5%), consisting of *avoid or stop debates* (23.8%), *share less* (21.5%), *distance from community* (12.3%), *decrease engagement* (9.2%), and *leave parts of the community* (3.8%). The second dimension was *avoid interpersonal risks* (20.0%), including *self-censorship*

(11.5%), *avoid situations* (7.7%), and *stop asking questions* (3.1%). A third dimension, *emotional response* (16.9%), captured *self-directed emotions* such as loneliness and anxiety (9.2%), *other-directed emotions* such as disappointment and annoyance (6.9%), and *loss of confidence* (3.1%).

Two smaller, but conceptually important, dimensions related to more constructive responses. *Improve situation* (10.8%) included *improve safety for others* (5.4%) and *focus on the positive* (6.9%), while *personal reflections* (8.5%) captured *re-evaluate expectations* (3.1%), *recognize larger implications* (3.1%), and related reflections (2.3%). Finally, 11.5% of participants reported *no effect*.

Overall, the dominant pattern is withdrawal: when psychological safety is low, many practitioners reduce their contributions, avoid situations where they might be exposed, or disengage from parts of the community. Alongside this behavioral withdrawal, participants frequently report negative emotional consequences. These patterns are consistent with the conceptualization of psychological safety as reducing the perceived interpersonal cost of speaking up: when that cost is high, contribution and risk-taking decline.

4.5.5. Group differences in qualitative responses

We now consider how these qualitative experiences vary across demographic and professional subgroups. Tables 8 and 9 report, for each subgroup (gender, age, seniority, content creatorship, and role), the percentage of participants associated with each aggregate dimension identified in Sections 4.5.3 and 4.5.4. Most group differences are not statistically significant. However, men report a significantly lower percentage of *negative interaction patterns* (24.7%) than women (47.5%) and other genders (55.6%). More pronounced differences appear across age groups: the 46-55 cohort reports the highest proportion of *exclusionary behaviors* (80%), the 65+ cohort reports the highest proportion of *negative interactions* (75%), and the 26-35 cohort reports the highest proportion of *hostile behaviors* (58.3%).

These results should be interpreted with caution, as subgroup sizes vary and can be small (between 4 and 51 participants; see Table 1). Nonetheless, they are broadly consistent with the quantitative results reported earlier, where few significant subgroup differences were found, aside from some effects for age and role in online settings. A reasonable conclusion is that, while certain groups may be more likely to notice or report specific patterns, the overall qualitative experience of psychological safety is relatively similar across demographic and professional subgroups.

Table 8: Group Differences in Behaviors Reducing Psychological Safety and Effects on Participants by Gender, Age, and Content Creation (%, N=143)

	Gender				Age (years)						Content Creator	
Dimension	Women	Men	Other	p	26-35	36-45	46-55	56-65	66+	p	Yes	No
Behaviors Reducing Psychological Safety												
Negative interactions	47.5	24.7	55.6	.016*	–	41.7	39.0	18.0	75.0	.003*	34.6	32.7
Hostile behaviors	32.5	33.3	33.3	.996	58.3	–	24.4	44.0	–	.011*	38.3	24.5
Exclusionary behaviors	65.0	63.0	55.6	.817	58.3	51.2	80.0	57.9	–	.006*	67.9	55.1
Unsafe environment	7.5	8.6	–	.937	–	25.0	–	9.8	–	.113	9.9	6.1
No experience	10.0	8.6	–	.952	–	9.8	–	21.1	–	.010*	11.1	6.1
Effects of Low Psychological Safety												
Reduce contributions	60.0	58.0	55.6	.963	50.0	63.4	58.0	68.4	–	.397	54.3	65.3
Avoid interpersonal risks	25.0	18.5	–	.560	25.0	–	22.0	18.0	–	.560	19.8	20.4
Emotional response	20.0	16.0	–	.772	–	19.5	18.0	–	–	.884	14.8	20.4
Personal reflections	10.0	6.2	–	.242	–	12.2	6.0	–	–	.686	7.4	10.2
Improve situation	17.5	6.2	–	.087	–	7.3	12.0	–	–	.808	9.9	12.2
No effect	10.0	12.3	–	.931	–	17.1	6.0	–	–	.402	12.3	10.2

* indicates $p < .05$. Cells with fewer than 3 participants suppressed (–). Welch ANOVA with Games-Howell post hoc.

Table 9: Group Differences in Behaviors Reducing Psychological Safety and Effects on Participants by Seniority and Role (%, N=143)

	Seniority (years)				**Role**						
Dimension	2-5	5-10	10+	*p*	Scrum M.	Agile C.	Trainer	Consult.	Other	Oth/PNS	*p*
Behaviors Reducing Psychological Safety											
Negative interactions	42.9	32.7	32.3	.845	45.5	30.4	40.0	21.1	20.0	–	.106
Hostile behaviors	28.6	28.6	38.5	.498	27.3	32.6	46.7	21.1	40.0	–	.198
Exclusionary behaviors	64.3	55.1	69.2	.474	60.6	63.0	66.7	52.6	73.3	–	.725
Unsafe environment	–	10.8	–	.106	–	6.5	–	21.1	–	–	.276
No experience	6.1	6.1	9.2	.178	15.2	8.7	–	–	–	–	.668
Effects of Low Psychological Safety											
Reduce contributions	57.1	49.0	66.2	.331	51.5	58.7	66.7	52.6	73.3	–	.471
Avoid interpersonal risks	35.7	22.4	15.4	.300	12.1	26.1	–	36.8	–	–	.142
Emotional response	–	20.4	13.8	.486	24.2	13.0	26.7	–	–	–	.285
Personal reflections	–	10.2	6.2	.165	12.1	8.7	–	–	–	–	.165
Improve situation	–	10.2	9.2	.319	21.2	8.7	–	–	–	–	.079
No effect	21.4	12.2	7.7	.160	18.2	8.7	–	15.8	–	–	.505

No significant differences found ($p > .05$). Scrum M.=Scrum Master (n=35), Agile C.=Agile Coach (n=49), Trainer=Trainer/Facilitator (n=17), Consult.=Consultant (n=21), Other (n=18), Oth/PNS=Other roles/Prefer not to say (n=3). Cells with <3 participants suppressed (–).

4.5.6. Factors contributing to lowered psychological safety

The previous sections addressed differences in psychological safety between online and face-to-face interactions and across subgroups. We now turn to the factors that practitioners associate with lowered psychological safety in their professional learning interactions. Participants were asked to describe what, in their experience, reduces psychological safety. Using the Gioia methodology described in Section 4.3, we developed a data structure of first-order concepts, second-order themes, and third-order aggregate dimensions (Figure A.5 in Appendix A).

Four aggregate dimensions emerged from the qualitative coding. The most frequently reported dimension was *negative interaction patterns* (53.8%), consisting of the themes *negative communication* (33.1%), *bullying and aggression* (23.1%), *lack of empathy* (12.3%), and *personal factors* (3.8%). The second most frequent dimension was *tribalism* (52.3%), which grouped the themes *rigid views* (43.8%) and *exclusion* (18.5%). Next, 23.8% of cases contained codes related to the dimension *community dynamics*, comprising *competition* (17.7%), *online dynamics* (6.2%), and *lack of support* (2.3%). Finally, 5.4% of respondents described factors related to the dimension *leadership and governance*, including *lacking governance* (3.8%), *lack of positive modelling* (1.5%), and *leaders act badly* (1.5%).

In summary, participants predominantly associate lowered psychological safety with the style and quality of day-to-day interactions, particularly hostile or dismissive communication, and with "tribal" dynamics that signal closed groups and rigid positions. Governance and leadership are mentioned less frequently, but when they are, participants point to the absence of clear norms and visible role-modeling as amplifying the impact of negative interactions.

4.5.7. Strategies to improve psychological safety

We next examine strategies that participants believe could enhance psychological safety. Participants were asked to propose actions that might improve psychological safety for themselves and others in these learning interactions. As in the previous subsection, we applied the Gioia methodology (Section 4.3) to derive first-order concepts, second-order themes, and aggregate dimensions, which are presented in Figure A.6 in Appendix A.

Five aggregate dimensions emerged from the qualitative coding. The most frequently reported dimension was *improve interactions* (33.1%), which includes the themes *develop soft skills* (14.6%), *change personal mindset*

(13.1%), *communication and interaction strategies* (9.2%), and *build awareness of psychological safety* (5.4%). The second most frequent dimension was *improve dynamics* (30.0%), comprising *community norms* (15.4%), *stand up to norm violations* (6.9%), *diversity and inclusion* (6.2%), and *structural changes* (18.5%). A further 20.8% of cases contained codes in the dimension *leadership and moderation*, which includes *lead by example* (13.1%), *moderation* (6.9%), and *leadership style* (1.5%). The dimension *personal coping* (17.7%) captures strategies such as *acceptance* (2.3%) and treating psychological safety partly as a *personal responsibility* (2.3%). Finally, 17.7% of participants reported that they *do not know* how psychological safety could be improved.

Taken together, these results suggest that practitioners see a shared responsibility for psychological safety. Many emphasize individual-level skills and mindsets that support respectful and constructive disagreement, while also highlighting the role of explicit norms, moderation, and structural arrangements that discourage hostile behavior. At the same time, the sizeable "do not know" category indicates that translating the abstract idea of psychological safety into concrete, actionable practices remains challenging for a substantial subset of practitioners.

5. Phase II: Member checking

The first phase of this study evaluated psychological safety in face-to-face and online learning interaction modalities, and explored perceived drivers of lower psychological safety and strategies to enhance it. Because psychological safety is inherently subjective, we sought to validate and refine these findings by engaging in member checking during Phase II. Member checking is a method in which researchers share synthesized findings with participants to assess credibility and obtain feedback [54]. Given the collaborative focus of this research, it also allowed us to draw on practitioners' expertise in interpreting the results and formulating new questions. Accordingly, we adopted the structured member-checking approach described by McKim [59].

5.1. Participants

Data for Phase II were collected through an online survey deployed using IBM Qualtrics between June 2024 and July 2024. As in Phase I, we used a purposive non-probabilistic sampling strategy [3]. Because Phase I data collection was anonymous, prior participants could not be contacted individually; instead, the Phase II survey was promoted in the same professional communities as Phase I. A screening question asked whether respondents had also taken part in Phase I.

The Phase II survey combined Likert-scale items with essay questions, as described in Section 5.2. The full instrument is provided in Table A.26 in Appendix A. Following McKim's recommendations [59], we did not ask participants to review raw data, which can be burdensome and yields limited feedback for larger datasets [90]. Instead, the survey presented key Phase I results using visualizations (e.g., histograms and bar charts) and short, factual statements based on preliminary analyses, for example, "Psychological safety is high in face-to-face interactions (3.88 out of 5)" and "Psychological safety is low in online interactions (2.91 out of 5)". Participants were asked to indicate the extent to which these findings matched their expectations and to offer interpretations and reflections in open-text responses. A screenshot is shown in Figure 2. The results reported for face-to-face and online psychological safety differed slightly (-.08 and -.09 respectively) from those of the present study, as subsequent, more rigorous SEM analyses indicated that two items in the Psychological Safety scale showed poor model fit and were therefore excluded as described in Section 4.4.1. This modification did not affect the conclusions reported in Phase II.

A total of 31 respondents completed the survey. Because recruitment relied on open community channels, a precise response rate cannot be computed. A data screening identified one careless response with largely empty or non-substantive answers [60], which was removed. The final sample ($N = 30$) is summarized in Table 10.

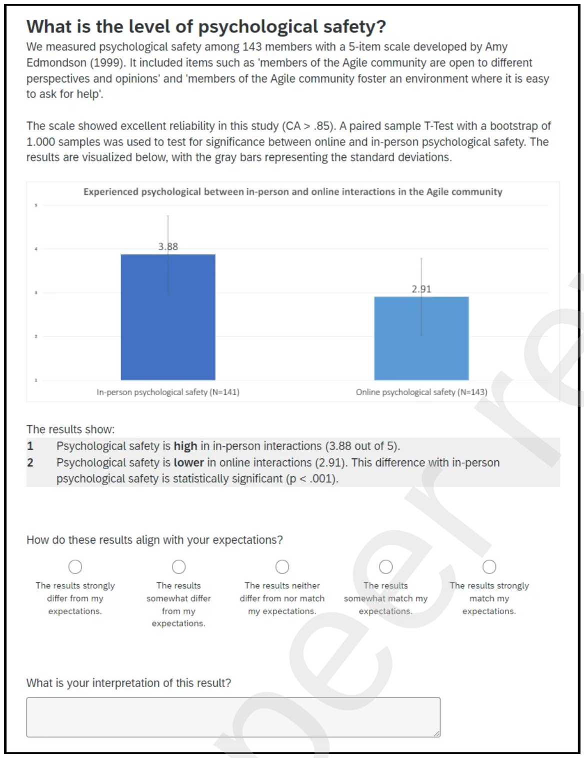

Figure 2: Example of one step in the survey for Phase II

Table 10: Sample Composition for Phase II

Variable	**Category**	N	%
Total respondents		30	100.0
Participated in Phase I			
	Yes	8	26.7
	No	15	50.0
	Do not remember	7	23.3

5.2. Measurements

Alignment of results with expectations by participants: The results from the quantitative and qualitative measures in Phase I were presented divided across discrete steps in the survey, each with a 5-point Likert item to assess the degree to which these results matched the expectations of the participant ("How do these results align with your expectations?"). Detailed breakdowns from Phase I were omitted to avoid overloading participants. Alignment was evaluated for four areas of the results: psychological safety between online and face-to-face learning interaction modalities, gender differences, role differences, and age differences. Seniority and content creatorship were omitted for brevity.

Interpretation of results: To collect interpretations from participants, one essay question was included in each step of the survey that presented results: "What is your interpretation of this result?". Additionally, one essay question was included at the end of the survey to collect follow-up questions: "Based on the results you've interpreted so far, what follow-up questions come to mind for you?". This qualitative data was coded as described in Section 5.3.

Interest in research process: This was assessed with a 5-point Likert item ("To our knowledge, this is the first time the community actively contributes to the sense-making of scientific evidence. Did you find this process interesting?").

Participation in Phase I: Near the end of the survey, participants were asked if they remembered participating in Phase I. A single categorical item was used - "Did you participate in the survey on psychological safety yourself?" - with the options "Yes", "No", and "Do not remember".

5.3. Analysis

The quantitative data was analyzed using IBM SPSS. Outlier detection was performed with stem-and-leaf plots. While some outliers were present on both sides of the distribution, we retained them as they seemed genuine cases. No data was missing, so all records were included in the analyses. The distributions of our continuous measures were significantly non-normal, as indicated by Kolmogorov-Smirnov tests, although kurtosis (< 3) and skewness (< 2) remained below their recommended thresholds [39]. The assumption of equal variance was not violated, as Levene's test returned a non-significant result for each of our continuous measures. To reduce bias in parameter estimates due to non-normality and small subgroups, a nonparametric Kruskal-Wallis independent samples median test was applied to compare medians across subgroups based on prior participation in Phase I.

The first author coded qualitative responses into broad themes.

We performed a post hoc power analysis using G*Power [27], version 3.1.9. We determined that our sample for Phase II allows us to capture small effects

($d = .020$) with a power of 60% ($1 - \beta = .600$) and 87.4% ($1 - \beta = .874$) for moderate effects ($d = .050$). Thus, we are confident that we can detect most moderate effects.

5.4. Results

The results for the quantitative measures are reported in Table 11. No significant differences were found based on whether participants also participated in Phase I, indicating no bias due to prior participation.

Table 11: Alignment of Results with Participant Expectations in Phase II

	All ($N = 30$)		**Phase I** ($N = 8$)		**Not Phase I** ($N = 15$)		**Unknown** ($N = 7$)		
Variable	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*p*
Alignment with participant expectations									
PS: Face-to-face vs. online	4.167	0.913	4.375	1.061	4.000	0.926	4.286	0.756	0.175
PS: By gender	3.933	0.828	3.875	0.991	3.800	0.775	4.286	0.756	0.698
PS: By role	3.333	1.061	3.375	1.061	3.267	1.100	3.429	1.134	0.587
PS: By age	3.067	1.258	3.000	0.926	3.000	1.464	3.286	1.254	0.310
Evaluation									
Interest in research process	4.233	0.679	4.375	0.744	4.133	0.640	4.286	0.756	0.407

Note. M = mean; SD = standard deviation; PS = psychological safety. Group differences tested using Kruskal-Wallis non-parametric ANOVA. No significant differences emerged between groups at $\alpha = .05$.

5.4.1. Evaluation by participants of difference between face-to-face and online interactions

The level of psychological safety reported in Phase I, as well as the difference between online and face-to-face interactions, matched the expectations of most participants ($M = 4.167, SD = .913, N = 30$). One participant noted "I feel safer to be open and honest when meeting in person vs online". Another wrote *"This does not surprise me, I suspect that you see this difference between in-person and social media engagement in general"*. However, some participants expected larger differences: *"I would expect in-person to score higher and online to score a lot lower"*. Table 12 reports four Hypotheses generated by participants. Seven participants suggested that the lack of social cues and body language reduces the potential for psychological safety in online interactions (i.e. *"in-person interactions provide more social cues to* *support understanding"*). Seven participants hypothesized that online interactions typically involve more people who are unfamiliar with each other, and those weaker social bonds limit psychological safety (i.e. *"I assume that with local meetups people already have a relationship with their peers.*). Third, participants offered that people in online interactions are less willing or capable of showing empathy, making their responses less safe (i.e. *"It seems people naturally care less about being polite and thoughtful when disagreeing [in online interactions]"*). Finally, one participant pointed out that online interactions may be more ad-hoc and less facilitated, with fewer opportunities to build psychological safety.

Table 12: Participant-Generated Hypotheses: Face-to-Face vs. Online Psychological Safety Differences

Hypothesis	n
Lack of social cues and body language in online interactions makes it harder to build psychological safety	7
In online interactions, people have weaker social bonds or personal connections	7
People are less capable or willing to show empathy in online interactions	6
Online interactions are less prepared and facilitated	1
Other	7

Note. $N = 28$. Participants could be coded into one or more categories.

Table 13: Participant-Generated Hypotheses: Gender Differences in Psychological Safety

Hypothesis	n
Gender bias and male dominance in communities at large	10
Men are less sensitive to situations of low psychological safety	4
Men are more self-confident than other genders	2
Gender is less visible in online interactions	1
Other	7

Note. $N = 24$. Participants could be coded into one or more categories.

Table 14: Participant-Generated Hypotheses: Role Differences in Psychological Safety

Hypothesis	n
Level of personal interest in creating psychological safety varies by role	9
Proneness to criticism due to role (e.g., trainers, consultants)	5
Certain roles spend more time with groups, thus allowing for more safety to form	2
Certain roles may be accompanied by higher need for ego confirmation (e.g., trainer/facilitator)	2
Other	8

Note. $N = 26$. Participants could be coded into one or more categories.

Table 15: Participant-Generated Hypotheses: Age Differences in Psychological Safety

Hypothesis	n
Generational differences in sensitivity to safety and unsafety	11
Generational differences in how we interact online shape how we respond to it	5
Younger age groups are correlated with other types of roles	1
Other	9

Note. $N = 26$. Participants could be coded into one or more categories.

5.4.2. Evaluation by participants of gender difference for psychological safety

Gender differences were mainly as expected by participants for overall psychological safety ($M = 3.933, SD = .828, N = 30$). The participants generated four discrete hypotheses, listed in Table 13. Ten participants ventured that the qualitative results reflected the gender bias and male dominance in communities at large (i.e. "*Matches both my experience in society and in online interactions per se as a woman*"). Four participants attributed the differences to a lower sensitivity to their psychological safety and that of others in men compared to other genders (i.e. *"Men may not be aware that those of other genders feel uncomfortable speaking up and are more likely to feel safe themselves."*). A third, related hypothesis offered by two partici-

Table 16: Themes in Participant-Generated Follow-Up Research Questions

Theme	n
Comparison to other or more general populations	4
Inquiries or feedback on Phase I	4
How to improve safety (e.g., how to decrease dogmatism, how to improve online, role of leadership)	3
Other measures of psychological safety and correlations with outcomes (e.g., performance)	2
Investigate role of potential factors (e.g., kind of interaction, social factors)	2
Other	6

Note. $N = 18$. Participants could be coded into one or more categories.

pants was that men typically act more self-confident, which makes them less sensitive. Finally, one participant noted that gender differences may be less salient in online interactions, which explains the small differences reported.

5.4.3. Evaluation by participants of role difference for psychological safety

The effects of role on psychological safety ($M = 3.333, SD = 1.061, N = 30$) aligned less with expectations. Participants broadly generated four discrete Hypotheses, listed in Table 12. Nine participants speculated that roles vary in terms of personal stake in creating psychological safety, and are thus more sensitive to it (i.e. "*As a Scrum Master, I often talk about psychological safety with my teams and others around my team. Will that make me more likely to feel safe myself [...]?*"). Five participants hypothesized that some roles invite more criticism, such as trainers and consultants, who are typically seen as bringing solutions. Another potential factor is that some roles typically spend more time with the same people, and are thus more likely to establish psychological safety (i.e. "*Scrum Masters may work with the same groups more frequently than trainers*"). Finally, two participants speculated that higher needs for ego confirmation accompany some roles ("*Perhaps it is a challenge to the ego of trainer/facilitator.*").

5.4.4. Evaluation by participants of age difference for psychological safety

The differences in psychological safety by age ($M = 3.067, SD = 1.258, N = 30$) were least similar to what participants expected. Three discrete hypotheses were extracted from the responses, listed in Table 15. Eleven participants speculated that generational differences would make some people more sensitive to their psychological safety and/or that of others (i.e. "*is it so that the older you get, the safer you feel due to your level of maturity?*"). Related to this, five participants speculated that different generations have different levels of experience with online interactions, i.e. "*Younger groups [are] very much used to not taking everything serious on social media, [whereas] the oldest people [are] leaning on their experience. Middle-aged [are] having the most troubles?*". Finally, one participant noted that the reported effects may be due to younger generations having different roles, and different levels of exposure to psychological (un)safety.

5.4.5. Follow-up research questions

At the end of the survey, participants were also asked to generate more follow-up questions based on the results. Five discrete themes were extracted from the responses, listed in Table 16. Most participants were curious about how the results compared to other or larger populations (i.e. "I feel that the problems we have in our community exist on a broader scale, so curious what we could do about this in such a large group"). Several participants also made suggestions to improve Phase I, particularly some definitions related to Agile terminology. Three participants also wondered how to improve psychological safety, as well as the specific factors contributing to it. Other participants were interested in different ways to measure psychological safety and how it correlated with (team) performance (i.e. "Do you know the connection between psychological safety and performance?"). Finally, two participants suggested exploring certain factors, specifically social factors, and how the type of interaction shapes psychological safety.

5.4.6. Interest in the research process

Finally, participants were asked to rate how interesting they found the process of reviewing Phase I results and generating ideas. Interest was high ($M = 4.233, SD = .679, N = 30$). Participants described how reviewing the results was useful, with quotes such as "*It's fun and helps with self-reflection*", "*Very interesting - please continue*" and "*Makes me feel valued and valuable*". One participant was more skeptical and wrote "*Everyone will respond to surveys differently based on their context. I do not feel it really rendered realistic results. Despite it being anonymous*". Several participants noted that while the process was valuable and engaging, the amount of data was quite overwhelming and could have been grouped more effectively. Two participants also noticed two mistakes in the visualizations in the results.

6. Discussion

This study examined how software practitioners experience psychological safety in learning interactions in online and face-to-face interaction modalities. Using a two-phase mixed-method design and member checking, we involved practitioners both in generating data and in interpreting the findings. In Phase I, 143 participants reported substantially lower psychological safety in online interactions than in face-to-face ones, with no significant differences between demographic variables of gender, seniority, age, role and content creatorship. Qualitative data indicated that exclusionary behaviors and negative interactions are most common, often leading to reduced contributions and negative emotional responses. Moreover, group differences between the qualitative experience of psychological safety were found between genders and age groups. Suggested improvement strategies centered on developing soft skills, establishing explicit norms, and ensuring supportive leadership and moderation. In Phase II, participants validated the key patterns and offered reflective commentary, confirming the relevance and credibility of the results for their own practice.

6.1. Factors contributing to low psychological safety

Our results show that psychological safety in learning interactions in professional software engineering communities is most often eroded by how people communicate with one another, rather than by disagreement per se. The most frequently reported contributors were exclusionary behaviors (63.1%) and negative interaction patterns (33.8%), including bullying, personal attacks, sarcasm, dismissiveness, and other forms of hostile or insensitive communication. Such behaviors directly increase the perceived interpersonal cost of speaking up and are particularly impactful in online settings, where nonverbal cues and opportunities for informal repair are limited [50, 41, 93, 98]. Moreover, participants typically responded to low psychological safety in any modality by withdrawing their contributions (58.5%), avoiding interpersonal risks (20.0%) and emotional responses (16.9%), thus reducing learning potential.

Phase II adds an interpretive layer to these findings. During member checking, practitioners did not merely recognize exclusionary or hostile behaviors; they actively reasoned about why such behaviors are more prevalent or more damaging in online learning settings. Participants attributed lowered psychological safety to structural properties of online interaction, including reduced social cues, weaker interpersonal bonds, limited facilitation, and diminished opportunities for empathy. This sense-making shows that practitioners recognize that psychological safety is moderated by the structure and design of interaction contexts.

Given that psychological safety is much lower in online settings, combined with the increasing need for continuous learning in such settings, the central challenge becomes to improve the quality of online learning interaction. While technical solutions such as emojis or reaction icons attempt to convey tone, prior work suggests they also introduce ambiguity [25, 85]. By contrast, interventions that focus on interpersonal and communication skills—such as awareness of how feedback is phrased, how humor is used, or how disagreement is framed are more directly aligned with the interpersonal risk at the heart of psychological safety. Our findings contribute a receiver-centric perspective on these behaviors, complementing observational taxonomies of psychological safety–related behaviors such as voice, silence, supportiveness, and familiarity [67]. Making these behavioral categories explicit helps practitioners recognize not only which interaction patterns are experienced as unsafe, but also how their effects may be amplified by online modalities.

Tribalism was another prominent theme. Participants described rigid group boundaries, "camps", and exclusionary practices that made some voices feel less legitimate. Social Identity Theory [95] offers a useful lens: in-group identification and out-group differentiation can increase cohesion for some, yet simultaneously discourage participation by others. These dynamics can be amplified online through group polarization, self-censorship, and groupthink [58, 45, 68]. Although such tendencies are deeply rooted [32], they can be moderated by establishing shared norms [10], clarifying goals and expectations [70], and modeling inclusive behavior through leadership [52].

Structural and facilitation practices also matter. Digital platforms can support psychological safety through transparent moderation, clear codes of conduct, and interaction formats that distribute voice more evenly. Facilitation methods such as Liberating Structures [56, 48, 86, 28, 77] offer practical scaffolding by providing structured interaction formats. Our findings suggest that these approaches are promising candidates for further study in software practice.

6.2. Moderators of psychological safety

Consistent with prior work [105], we observed no significant differences in reported levels of psychological safety across demographic variables. However, the qualitative findings indicate that demographic groups differ in which behaviors they experience as threatening. In particular, women and participants identifying as other or unknown genders reported substantially more negative interaction patterns than men. Differences also emerged across age groups: participants aged 26–35 reported the highest incidence of hostility, while those aged 46–55 most frequently reported exclusionary behaviors. Participants aged 56 and above reported comparatively few negative interaction patterns, yet still described notable experiences of exclusion.

Phase II provides important insight into how practitioners themselves interpret these differences. Rather than viewing demographic patterns as fixed or deterministic, participants framed them in terms of differential sensitivity, exposure, and socialization into online communication norms. Several participants suggested that generational experiences with digital interaction shape how safety threats are perceived and navigated, an interpretation that aligns with existing work on generational communication styles [2]. These reflections suggest potential generational differences in how online interactions are perceived and managed. Younger cohorts may be more attuned to tone, pacing, and public visibility in online discourse, whereas older cohorts may be more detached from interactional dynamics or less affected by them. Participants in intermediate age groups, often described as "digital migrants", may experience greater friction when navigating between offline norms and online expectations [63].

Phase II responses also surfaced tensions in how psychological safety itself is understood. Some participants expressed concern that heightened sensitivity could constrain open conversation, invoking notions such as "cancel culture." Others emphasized that psychological safety is partly an individual responsibility rather than a collective condition. These tensions reflect a common misconception that psychological safety entails the avoidance of conflict. As Edmondson [23] emphasizes, psychological safety does not suppress disagreement; rather, it enables individuals to engage in dissent and debate without fear of social or reputational harm. The fact that these distinctions were actively articulated by participants during member checking suggests that practitioners are not only subjects of psychological safety dynamics, but capable interpreters of them.

6.3. Psychological safety, toxicity and incivility

Psychological safety is typically operationalized in the context of a bounded group with stable membership and task interdependence and emphasizing behaviors such as admitting mistakes, experimenting with new ideas, and challenging established practices within ongoing work teams [21, 24]. We operationalized psychological safety in the context of learning interactions, which are more episodic, often involve participants without prior relationships, and occur in contexts with fluid or ill-defined boundaries, asymmetric expertise, and limited social cues, particularly online [103, 93]. Like Edmondson [21], we included the ability to ask for help and offer different opinions, but de-emphasized task-related behaviors such as admitting mistakes and experimentation. At its core, both operationalizations are concerned with the perceived interpersonal cost of speaking up. Indeed, the concerns of participants focused on dismissiveness, lack of responsiveness, and exposure to ridicule or exclusion, which suggests that psychological safety in loosely bounded communities is closely tied to inclusion and social standing [51, 65]. Phase II reinforced this contextualization. Practitioners repeatedly framed psychological safety as a condition that enables critical debate, not one that limits it. This practitioner-led interpretation strengthens the argument that psychological safety should be understood as context-sensitive, with behavioral indicators that depend on the nature of participation and interaction [67].

This operationalization also clarifies the relationship between psychological safety and adjacent constructs such as toxicity and incivility, which are frequently studied in software engineering communication. Toxicity refers to high-intensity behaviors with a clear intent to harm or marginalize others, including swearing, rudeness, and abusive language [61, 80, 79]. Incivility, by contrast, encompasses low-intensity behaviors with ambiguous intent, such as dismissiveness or a lack of respect [83]. Many of the threats to psychological safety described by participants in this study fall within these categories. Psychological safety, however, concerns the anticipation of encountering such behaviors when engaging in interpersonal risk-taking.

To date, research on toxicity and incivility in software engineering has primarily focused on their (automated) detection [79, 80], particularly in open-source communities [13, 61] and code review contexts [29]. Our study extends this work by documenting manifestations of toxicity and incivility in learning interactions beyond these settings and linking them directly to participants' perceptions of psychological safety. We show how such behaviors erode psychological safety, leading to reduced participation, emotional distress, and avoidance of interpersonal risk. Finally, our findings point to strategies for protecting and restoring psychological safety, not only by limiting toxic and uncivil behavior, but also by restructuring interactions, reducing tribalism, and fostering effective leadership.

6.4. Recommendations for Practice

This study highlights several actionable insights for professionals engaging in learning interactions in the professional community, as well as those designing, facilitating and leading them. We now turn to a discussion of what practitioners can change tomorrow given these insights. A summary is provided in Table 17.

6.4.1. Improving interaction quality

Negative interaction patterns constituted the most frequently reported cause of threat behaviors (53.8%), while the development of appropriate soft skills to improve interactions was the most frequently reported improvement strategy (33.1%). The majority of threat behaviors in this category — including impoliteness, sarcasm, inappropriate humor, lack of respect, and unconstructive criticism — can be characterized as forms of incivility, or low-intensity behaviors with an ambiguous intent to harm [83]. While such behaviors may at times be deliberately harmful, in other cases receivers may infer harmful intent incorrectly due to individual biases or attribution errors. This risk is amplified in online settings, where interpretive cues are reduced and opportunities for clarification and interactional repair are limited [50, 41, 93]. Consequently, responsibility for maintaining constructive interactions is shared by both senders and receivers, each of whom must remain attentive to communicative ambiguity and actively seek to reduce it.

One way practitioners can address this challenge is by developing metacommunicative competence [102], defined as the ability to reflect on, and explicitly communicate about, the communication process itself. This includes practices such as verifying understanding (e.g., “Am I understanding you correctly?”), initiating repair when misunderstandings occur (e.g., “I see I misunderstood you. Let me to try to reformulate my view."), and engaging in perspective-taking to anticipate how a message may be interpreted by others. A playful strategy is to append a statement one wants to make with “, you idiot" as a thought experiment. If the statement still works, it needs to be softened until it does not. Metacommunicative competence further involves making communicative intent explicit at the outset, for example by signaling appreciation before offering critique, thereby reducing uncertainty regarding evaluative intent. More generally, it requires recognizing the high likelihood of ambiguity in online communication and adopting a cautious approach that incorporates explicit encouragement and appreciation.

Importantly, not all negative interactions are unintentional. When behaviors are perceived as intentionally harmful and attempts at repair are absent or unsuccessful, metacommunicative competence also entails articulating the experienced impact, setting boundaries, and, where necessary, disengaging from the interaction. In addition, practitioners play a collective role in reinforcing constructive interaction norms by supporting one another in calling out toxic or uncivil behavior and by upholding shared expectations regarding respectful communication.

Given the growing importance of online learning interactions in professional communities, organizations and educational institutions can play a proactive role by supporting the development of these skills. Leaders of learning communities may incorporate concrete examples of metacommunicative competence into codes of conduct and working agreements and model such practices in their own interactions. Furthermore, online platforms (e.g., professional social networks, collaboration tools, and discussion forums) can improve interactions by providing mechanisms to monitor interaction quality, prompt tone adjustments, and surface relevant community norms — potentially augmented through the use of AI-based tooling [64]. For example, Sharma et. al. [84] developed an AI agent that improved conversational empathy between peers by providing feedback during interactions.

6.4.2. Creating awareness of psychological safety

58.5% of participants in this study reduced their contribution in response to low safety, or withdrew altogether. But experiences of low psychological safety are not uniform. While the quantitative analysis did not reveal meaningful differences across gender, role, age group, or seniority, the qualitative data showed variation in what different groups experienced as threatening. At the same time, psychological safety is conceptualized as a shared belief within a group [21]. This underscores why learning communities should engage in explicit, collective sense-making about what psychological safety means in practice and how it can be fostered, a need that was also articulated by many participants.

This study's conceptual model, together with Edmondson's foundational work on psychological safety [21, 24, 22] and frameworks such as O'Donovan et al. [67], provides useful starting points for such efforts. Crucially, psychological safety should not characterized as a lack of conflict, but as shaping an environment in which disagreement can occur productively without jeopardizing individuals' social standing or willingness to participate. Learning communities can translate these insights into co-created norms regarding constructive disagreement, the inclusion of newcomers, or the interpretation of silence, while ensuring that these norms are consistently enacted in everyday interactions [52]. Prior work on the effectiveness of awareness-raising and norm-setting initiatives has shown mixed results, highlighting the need for continued empirical evaluation of how such interventions function in practice [82, 9, 20].

6.4.3. Improve community dynamics

In this study, 63.1% of participants identified exclusionary behaviors—such as dismissiveness and dogmatism—as a primary threat to psychological safety. Similarly, 52.3% of participants pointed to tribalism and rigid group boundaries as major contributing factors, alongside broader community dynamics (23.8%), particularly when these dynamics encourage competition. Although such dynamics are predictable from cognitive and social psychology [95, 58, 45, 68], they are not immutable and can be mitigated through deliberate structural interventions in how learning interactions are organized.

One promising strategy is to introduce interactional scaffolding that structures learning interactions in ways that promote active listening, diversity of perspectives, and more equitable participation [48]. A concrete example is the open-source collection of 33 Liberating Structures [56, 48], also cited by participants. Each structure provides a lightweight, purpose-driven interaction scaffold, including prompts, sequencing, and time boundaries, intended to guide group interaction without prescribing content. For instance, "1-2-4-All" supports idea generation and collective sense-making by sequencing individual reflection, paired discussion, small-group synthesis, and plenary sharing. This gradual increase in group size lowers the interpersonal risk associated with initial participation, distributes airtime more evenly, and reduces the influence of early or high-status speakers. Similarly, "Troika Consulting" structures peer consultation by assigning rotating roles of client and consultants within groups of three. Consultants discuss possible perspectives while the client listens without responding, followed by individual reflection. This role separation limits defensiveness and unproductive debate while enabling participants to receive diverse input in a psychologically safer manner.

Leadership was identified by 20.8% of participants as an important factor in improving community dynamics, particularly through moderation, role modeling, and the establishment of clear interaction norms. Leaders can help counter the formation of rigid "expert camps" by modeling openness to diverse perspectives, rotating roles and responsibilities, and intentionally pairing newcomers with more experienced members. Even relatively small, consistent leadership behaviors can signal inclusivity and legitimize participation from a broader range of contributors [103].

In predominantly text-based online settings, emerging evidence suggests that AI-supported tooling may further assist leaders and moderators by identifying potential instances of incivility and other threat behaviors for review, thereby supporting more timely and consistent moderation practices [29].

Table 17: Summary of recommendations for practice

Role	**Recommendation**
Interaction quality	
Individual practitioner	Practice metacommunicative competence by clarifying intent, checking understanding, initiating repair, and expressing appreciation.
Meetup facilitator	Model constructive communication, normalize clarification and repair, and intervene when behavioral threats appear in interactions.
Community organizer	Embed examples of constructive communication in codes of conduct, working agreements, and onboarding materials.
Tool / platform owner	Support interaction quality through detection of behavioral threats, tone prompts, and surfacing of community norms, potentially using AI.
Awareness of psychological safety	
Individual practitioner	Treat psychological safety as enabling constructive disagreement rather than conflict avoidance.
Meetup facilitator	Make expectations around disagreement, silence, and participation explicit during learning interactions.
Community organizer	Co-create, communicate, and consistently model shared norms for psychological safety.
Tool / platform owner	Increase norm visibility through onboarding cues, reminders and tone modulation.
Community dynamics and inclusion	
Individual practitioner	Support inclusion by amplifying diverse perspectives and resisting dismissiveness and dogmatism.
Meetup facilitator	Use interactional scaffolding (e.g., Liberating Structures) to equalize participation and promote active listening.
Community organizer	Design structures that counter exclusion, such as role rotation and newcomer integration practices.
Tool / platform owner	Enable inclusive interaction through collaboration affordances and AI-assisted moderation of behavioral threats.

6.5. Threats to validity

This section outlines the limitations of our study, guided by Russo et al. [72] and incorporating both qualitative and quantitative considerations. We address threats to credibility, transferability, dependability, and confirmability [38], along with internal, construct, external, and conclusion validity [104]. To promote transparency and reproducibility, we have made anonymized quantitative data, syntax files, and codebooks openly available on Zenodo[1].

Credibility & Internal. In qualitative research, credibility refers to the degree to which the findings accurately reflect the perspectives and experiences of participants [55]. We employed the Gioia framework [33] to rigorously structure and code the data through open and axial coding as part of our thematic analysis. Dual coding by two researchers yielded high inter-rater agreement ($\kappa > .90$), reducing interpretative bias. Additionally, the process was overseen by an expert researcher. See Section 4.3 for details and the replication package for the full codebook. Member checking was incorporated in Phase II, which confirmed broad recognition of the findings by participants, thereby reinforcing credibility [54].

Internal validity concerns the extent to which observed effects can be causally attributed [14]. We confirmed statistical assumptions [39] and applied robust methods, Structural Equation Modeling (Phase I), and Kruskal-Wallis median tests (Phase II), to mitigate distributional violations. Triangulation across qualitative and quantitative data sources further strengthened the robustness of inferences. While self-selection bias is a common limitation in voluntary surveys, we addressed this through anonymized participation, inclusion of opt-out options, and broad dissemination (Section 4). The variation in psychological safety scores, including 9.2% of respondents reporting no negative experiences, suggests sample diversity. A potential risk of collapsing heterogeneous settings and platforms into broad "online" and "face-to-face" modalities is that context-specific properties conflate with psychology. However, no significant differences in reported psychological safety where found between settings and platforms within modalities.

Common method bias caused by the use of a single instrument was controlled as recommended in statistical literature [69]. Recall bias was ad-

[1] A replication package is openly available under a CC-BY-NC-SA 4.0 license on Zenodo, DOI: `https://doi.org/10.5281/zenodo.18255132`.

dressed by asking participants to keep their most frequently-used platform in mind for online interactions. Without controlling for the recency of experiences, we can not rule out that more recent or very negative experiences influenced responses. On the other hand, this reflects a reality of psychological safety, or the expectation that interpersonal risk-taking will be punished based on prior experience - biased or not.

Transferability and external validity. Transferability and external validity concern the extent to which findings can be applied beyond the specific sample, settings, and time period studied [36]. Two limitations arise in our case. First, our sample consists of practitioners engaged professional learning settings related to Agile software engineering. The results may not transfer to software engineers broadly. Second, because we used purposive, non-probabilistic sampling, the sample is not statistically representative of all such communities or of the broader software engineering population.

We addressed these limitations in three ways. First, we provided detailed descriptions of the sample and context in Sections 4.1 and 5.1, enabling readers to assess the similarity between our setting and their own and to judge the transferability of the findings. Second, the community-level validation performed in Phase II showed that participants generally recognized and affirmed the patterns from Phase I, and there were no significant differences between respondents who had participated in Phase I and those who had not, which supports the robustness of the inferences within similar contexts. In addition to credibility, Phase II provided an opportunity to validate interpretations with participants, and to integrate participant reasoning into the discussion. Third, we promoted the study across multiple platforms, communities, and media, as described in Section 4, and engaged community leaders in disseminating the invitations, increasing the diversity of perspectives within the constraints of a purposive sampling approach.

Finally, psychological safety is often analyzed as a group-level construct using multilevel models in settings with stable, well-defined teams. In this study, participants engaged in episodic learning interactions across multiple, loosely bounded communities, making it infeasible to reliably nest individuals within identifiable groups. We therefore analyzed psychological safety as an individual-level perception of interaction contexts rather than as a shared group climate. As a result, our findings reflect perceived psychological safety in professional learning interactions and should not be interpreted as group-level climate effects. Future work could extend this research by collecting repeated measures within specific communities to enable multilevel analysis.

Dependability & Construct. Dependability in qualitative research refers to the consistency and reliability of findings throughout the research process [36]. Qualitative data was translated into insights through the Gioia methodology [33], which aims for scientific rigor in qualitative analyses. Two coders independently coded all cases for Phase I to reduce bias and iterated on a codebook through memoing. Inter-rater reliability indicated high agreement ($\kappa > .90$). The codebook and memos are available in the replication package.

Construct validity refers to the degree to which the measures used in quantitative analyses measure their intended constructs [14]. Psychological safety was measured with a scale that captured aspects of Edmondson's operationalization of psychological safety [21] deemed relevant to learning interactions in professional communities. Convergent validity was established by piloting the scale with a selection of high-loading items from Edmondson's original scale. The scale was applied twice, once for face-to-face and once for online interactions. A confirmatory factor analysis (CFA) confirmed that both measures represented distinct factors (see Table A.21 in Appendix A). The reliability for both measures exceeded the cutoff recommended in the literature ($CR >= .70$ [39]). Thus, we are confident that we reliably measured psychological safety in the context of online learning interactions.

Confirmability & Conclusion validity. In qualitative research, confirmability is the extent to which the findings are grounded in the data and not the researcher's biases or preferences [36]. To establish confirmability, we transparently documented our analysis process in this paper to allow traceability and auditing of our interpretations. We utilized open-source tools (QualCoder [18]) for our coding and provided the codebook and memos in the replication package.

Conclusion validity refers to the soundness of inferences about relationships among variables [15]. The Phase I sample was powered to detect moderate effects ($d = .50$) with 99.8% confidence. We applied appropriate statistical techniques per recommended guidelines [39] and triangulated data sources to ensure coherence of findings (Sections 4.3, 4.4 and 5.3).

Table 18: Summary of findings and implications

Finding	Summary	Implications
Lower safety online	Psychological safety was significantly lower in online interactions (M = 3.006, SD = 1.08, N = 143) than in face-to-face interactions (M = 3.965, SD = .92, N = 141).	Online settings provide fewer cues and fewer opportunities to repair misunderstandings. Organizers should explicitly design for psychological safety, co-create norms, and use facilitation practices that lower the interpersonal cost of speaking up.
Behavioral threats	Qualitative analysis (N = 120) identified exclusionary behaviors, negative interaction patterns, and hostility as primary threats, leading to reduced contributions, avoidance of interpersonal risks, and emotional strain.	Some behaviors have clear negative intent; many are ambiguous. Regardless of intent, widespread feelings of unsafety lead to withdrawal and reduced learning, making psychological safety a shared responsibility rather than only an individual interpretation issue.
Demographic differences	No significant effects emerged for gender, role, age, seniority, and content creatorship in the level of reported psychological safety, although some qualitative differences were found for age groups and genders.	Psychological safety is broadly similar across most demographics, but sensitivity and exposure vary. Behaviors that feel acceptable to some may feel unsafe to others. Guidelines should reflect this heterogeneity rather than assuming a uniform audience.
Strategies for improvement	Qualitative analysis (N = 111) highlighted three clusters of strategies: improving interactions (soft skills and communication), improving dynamics (norms and structure), and strengthening leadership and moderation.	Soft skills (e.g., active listening, inquiry, appreciative feedback) are central yet unevenly distributed. Organizers can lower the barrier by providing simple structures, prompts, and facilitation techniques that embed these skills into everyday interactions, especially online and in hybrid formats.

Note. M = mean; SD = standard deviation.

7. Conclusion

This study explored a paradox of online learning. While online environments are designed to broaden learner participation, they provide fewer means by which learners feel safe to take the interpersonal risks inherent to learning. We performed a two-phase mixed-method study that combined survey data, qualitative data and member checking to understand how psychological safety differs between face-to-face and online modalities, how it affects participants, what contributes to it and what can be done to improve it. As expected, psychological safety was substantially lower in online learning interactions than in face-to-face settings. While no significant differences were found in reported psychological safety by gender, role, seniority, age group and content creatorship, we did find notable differences in behavioral threats lower psychological safety by gender and age groups.

Thematic analysis showed that exclusionary behaviors, negative interaction patterns, and hostility are the most salient threats to psychological safety. Such behaviors are associated with reduced contributions, self-censorship, disengagement from parts of the community, and negative emotional reactions, all of which diminish the capacity of professional communities to support sustained learning. Practitioners highlighted four main levers for improvement: (i) developing interpersonal and communication skills, (ii) co-creating and upholding behavioral norms, (iii) using inclusive facilitation methods to distribute voice, and (iv) ensuring visible support and modeling from leaders and moderators. Together, these strategies offer practical guidance for organizers and platform providers seeking to foster more inclusive and learning-conducive environments, especially in digital settings.

We used member checking to validate our results with practitioners and actively engaged them in interpreting potential causes and consequences. Participants emphasized that psychological safety is shaped by contextual properties, and a condition to enable constructive disagreement without social or reputational harm.

This work advances research in three ways. First, it highlights the paradox of online learning and emphasizes the need for a better understanding to make learning interactions safer. Second, it proposes an empirically grounded model that links threat behaviors, perceived consequences, and contextual drivers. Third, it incorporates practitioner perspectives through member checking, aligning academic interpretation with lived experience.

Future work should examine whether these patterns hold beyond Agile-focused contexts and across different types of software engineering communities. Comparative studies across domains with varying cultures, levels of formality, and demographic compositions would help assess the generalizability of our findings and connect them to broader work on disruptive innovations in software engineering [92]. There is also a need for rigorous development and evaluation of targeted interventions, particularly those co-designed with practitioners and tailored to specific interaction modalities (e.g., synchronous versus asynchronous, text-based versus video, one-to-one versus many-to-many). Complementing self-report data with behavioral, interactional, or observational measures would support stronger causal inferences and contribute to more robust theory-building on psychological safety in software engineering. Taken together, these directions align with recent calls for human-centered approaches to software engineering and generative AI that foreground practitioners' experiences and social conditions [73, 71].

8. Acknowledgments

The authors would like to thank all practitioners and communities who contributed to Phase I and/or Phase II of this study.

9. Supplementary Materials

A replication package for the study is available at the following DOI: 10.5281/zenodo.18255132 under a CC-BY-NC-SA 4.0 license. The package includes SPSS syntaxes, statistical models used for Structural Equation Modeling, a codebook, and a coded dataset. Qualitative responses were omitted to protect the anonymity of participants.

10. Responsible disclosure

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Appendix A. Supplementary materials

Table A.19: Examples of thematic analysis: quotes, first-order concept, second-Order theme, and aggregate dimension

Quote	First-order concept	Second-order theme	Dimension
Experiences that reduced psychological safety			
"Ridicule when you do try to discuss (e.g., 'SAFe is bad and you're stupid to think otherwise')"	Ridiculing	Personal attacks	Hostile behavior
"Personal attacks, rather than raising questions about content."	Attacking the person	Personal attacks	Hostile behavior
"[People] get personal like 'it is obvious you have never done this"'	Gatekeeping	Dogmatism	Feeling excluded
"One upping on level of experience in response to posts."	One-upmanship	Posturing	Negative interactions
"Shooting down posts and comments of new or less experienced Scrum Masters in a firm manner."	Gatekeeping	Dogmatism	Feeling excluded
Effects of reduced psychological safety			
"I won't post my thoughts"	Contribute less	Share less	Reduce contributions
"Be careful in your expressing yourself"	No self-expression	Self-censorship	Avoid interpersonal risks
"I prefer not to express and share thoughts that I know to be controversial and able to cause poisonous reactions."	No dissenting views	Self-censorship	Avoid interpersonal risks
"Sometimes I feel very lonely"	Loneliness	Self-directed emotions	Emotional response
"I pull back from these communities as they become increasingly less valuable."	Leave community	Distance from community	Reduce contributions
Factors lowering psychological safety			
"Evangelistic, almost cultist, opinion about own ideas/methods/frameworks"	Dogmatism	Rigid views	Tribalism
"Behaving as if there are clubs that you are not 'allowed' to participate in, you don't deserve to have opinions."	Gatekeeping	Exclusion	Tribalism
"Emotionally charged language or behaviors"	Charged language	Bullying and aggressive behavior	Negative interaction patterns
"Leaders of the community show no respect to newcomers"	Don't welcome newcomers	Lack of positive modelling	Leadership and governance
"Constant sales pitches"	Self-promotion	Competition	Community dynamics
Strategies to improive psychological safety			
"Online forums could build their own working agreement, set boundaries for appropriate behavior and then uphold those boundaries."	Code of conduct	Community norms	Improve dynamics
"Always begin a comment with honest appreciation of something the other person said/wrote."	Show appreciation	Develop soft skills	Improve interactions
"Discussion panels that are intentionally a mix of those in the community and those outside discussing topics that matter to both"	Increase diversity	Diversity and inclusion	Improve dynamics
"I think groups should be moderated"	Moderation	Moderation	Leadership and moderation
"Talk about psychological safety more at the beginning of meetups"	Create awareness	Awareness of psychological safety	Improve interactions

Figure A.3: First-order concepts, second-order themes, and aggregate dimensions in the effects of low psychological safety as reported in Communities of Practice by participants ($N = 122$)

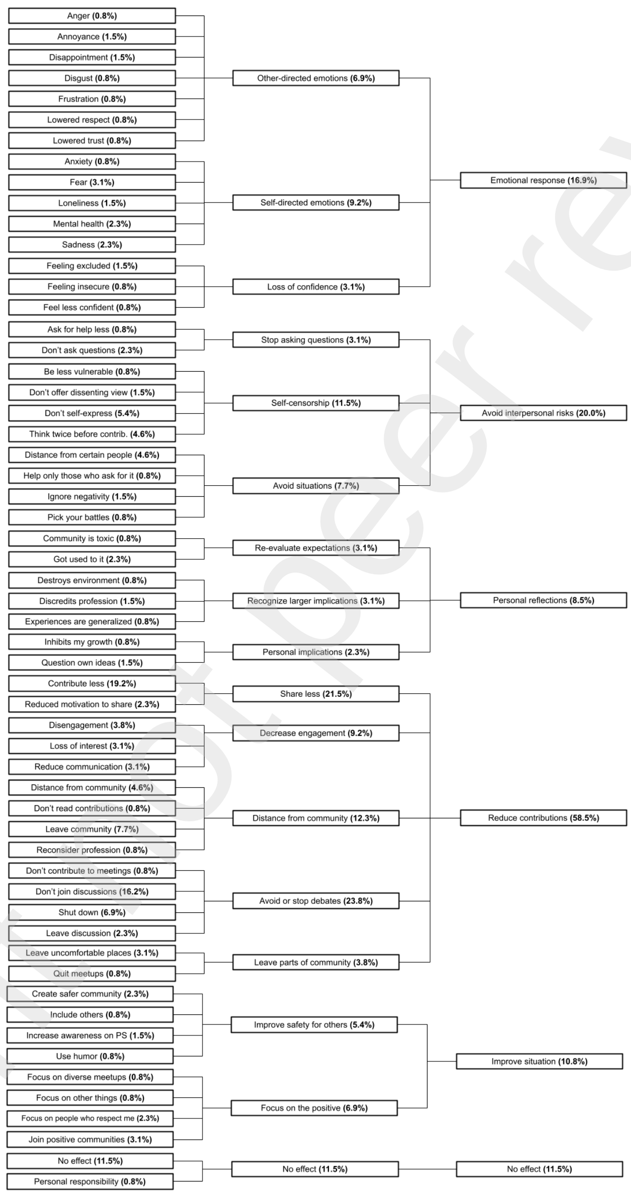

Figure A.4: First-order concepts, second-order themes, and aggregate dimensions in the behaviors that contributed to lowered psychological safety as reported in Communities of Practice by participants ($N = 120$)

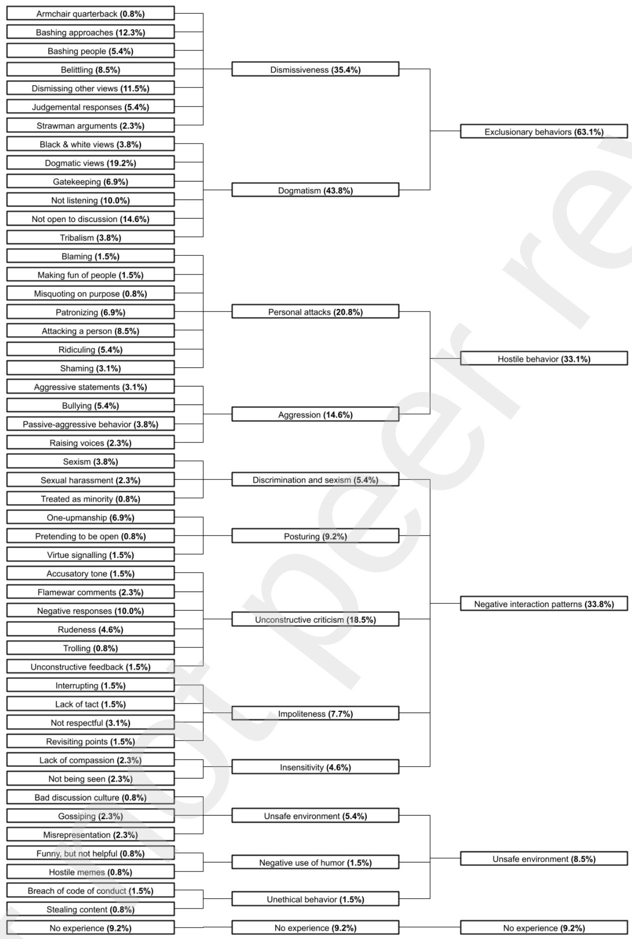

Figure A.5: First-order concepts, second-order themes, and aggregate dimensions in the factors contributing to lower psychological safety in Communities of Practice as reported by participants ($N = 118$)

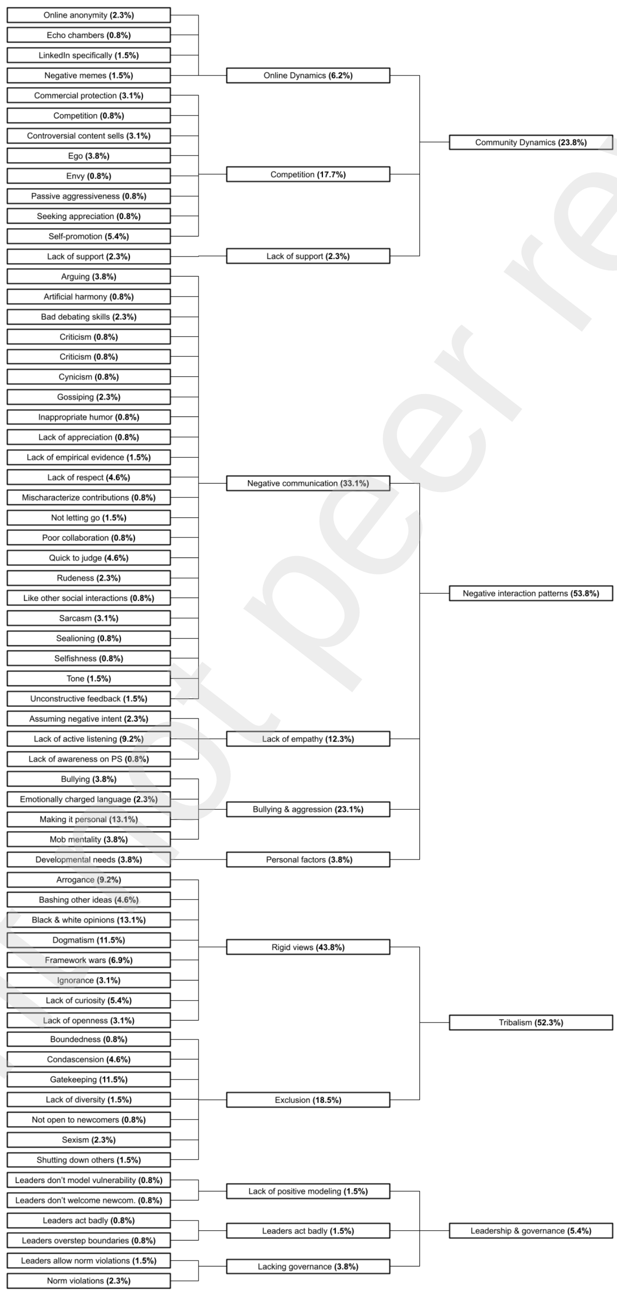

Figure A.6: First-order concepts, second-order themes, and aggregate dimensions in the strategies to improve psychological safety in Communities of Practice as suggested by participants ($N = 111$)

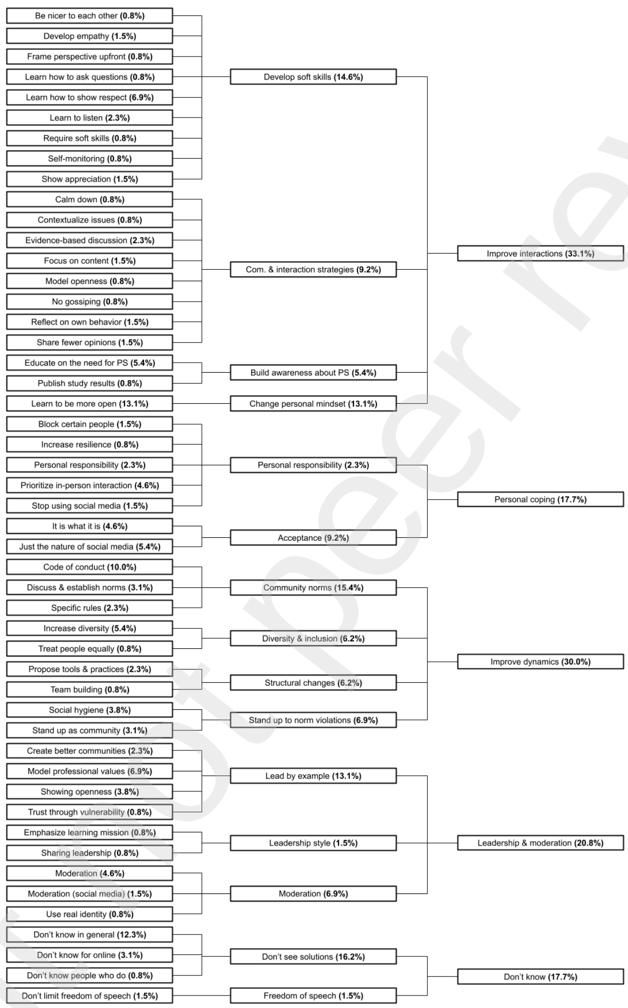

Table A.20: Descriptive statistics and Pearson correlations for continuous, ordinal, and control variables in Phase I. Correlations marked with an asterisk (*) are statistically significant at $p < .05$.

Variable	N	Mean	SD	Skewness	Kurtosis	1	2	3	4	5	6
1. Psychological safety (face-to-face)	141	3.96	0.92	-1.10	1.06	1.00					
2. Psychological safety (online)	143	3.01	1.08	-0.20	-0.87	0.43*	1.00				
Ordinal demographic variables											
3. Age group	143	3.82	1.13	0.69	0.72	-0.04	0.08	1.00			
4. Seniority in community	143	4.35	0.74	-0.98	0.57	-0.07	-0.06	0.38*	1.00		
Control variables											
5. Interaction frequency (face-to-face)	143	12.01	2.91	-0.98	0.46	0.09	0.02	-0.05	0.17*	1.00	
6. Interaction frequency (online)	143	13.63	4.66	1.05	1.79	0.07	0.12	0.14	0.19*	0.27*	1.00

Table A.21: Results of Confirmatory Factor Analysis from individual-level responses ($n = 143$). Principal Components Analysis with Oblimin rotation and Kaiser normalization. Items marked with asterisk (*) were removed in the final scale.

Item	1	2
Psychological safety 1 (Face-to-face)	.905	
Psychological safety 2 (Face-to-face)	.886	
Psychological safety 3 (Face-to-face)	.930	
Psychological safety 4 (Face-to-face)*	.592	
Psychological safety 5 (Face-to-face)*	.356	
Psychological safety 6 (Face-to-face)	.837	
Psychological safety 1 (Online)		.755
Psychological safety 2 (Online)		.814
Psychological safety 3 (Online)		.869
Psychological safety 4 (Online)*		.606
Psychological safety 5 (Online)*		.370
Psychological safety 6 (Online)		.877

Table A.22: Phase I questionnaire: online interactions and psychological safety

Variable	Item (stem in italics)	Type	Response options
How frequently do you share something with the Agile community on the following social media (e.g., comment, article, blog post, video)?			
Where do you meet (online)	LinkedIn	Categorical	Never; Less than once a month; At least once a month; At least once a week; At least once per day
Where do you meet (online)	Twitter / X	Categorical	(as above)
Where do you meet (online)	Medium	Categorical	(as above)
Where do you meet (online)	Mastodon	Categorical	(as above)
Where do you meet (online)	Reddit	Categorical	(as above)
Where do you meet (online)	Slack / Discord	Categorical	(as above)
Where do you meet (online)	Agile forums	Categorical	(as above)
Where do you meet (online)	YouTube	Categorical	(as above)
On the social medium you use most frequently, how do you experience the following?			
Psychological safety online #1	*On this medium, members of the Agile community foster an environment where it is easy to ask for help.*	Likert (1–5)	Strongly disagree; Somewhat disagree; Neither agree nor disagree; Somewhat agree; Strongly agree
Psychological safety online #2	*On this medium, members of the Agile community are open to different perspectives and opinions.*	Likert (1–5)	(as above)
Psychological safety online #3	*On this medium, people in the Agile community make an effort to understand each other's views.*	Likert (1–5)	(as above)
Psychological safety online #4	*On this medium, I frequently observe or experience behavior in the Agile community that lowers psychological safety.*	Likert (1–5)	(as above; reverse-coded)
Psychological safety online #5	*On this medium, it is important to me that interactions in the Agile community are psychologically safe.*	Likert (1–5)	(as above)
Psychological safety online #6	*On this medium, I experience psychological safety in the Agile community to be high.*	Likert (1–5)	(as above)

Table A.23: Phase I questionnaire: face-to-face interactions and psychological safety

Variable	Item (stem in italics)	Type	Response options
Where do you meet members of the Agile community face-to-face (video or in person)?			
Where do you meet (face-to-face)	Conferences	Categorical	Never; Rarely; Often; Very often
Where do you meet (face-to-face)	Online meetups	Categorical	(as above)
Where do you meet (face-to-face)	In-person meetups	Categorical	(as above)
Where do you meet (face-to-face)	Workshops or training	Categorical	(as above)
Where do you meet (face-to-face)	Other	Categorical	(as above)
During face-to-face interactions with members of the Agile community, how do you experience the following?			
Psychological safety face-to-face #1	*During face-to-face interactions, members of the Agile community foster an environment where it is easy to ask for help.*	Likert (1–5)	Strongly disagree; Somewhat disagree; Neither agree nor disagree; Somewhat agree; Strongly agree
Psychological safety face-to-face #2	*During face-to-face interactions, members of the Agile community are open to different perspectives and opinions.*	Likert (1–5)	(as above)
Psychological safety face-to-face #3	*During face-to-face interactions, people in the Agile community make an effort to understand each other's views.*	Likert (1–5)	(as above)
Psychological safety face-to-face #4	*During face-to-face interactions, I frequently observe or experience behavior in the Agile community that lowers psychological safety.*	Likert (1–5)	(as above; reverse-coded)
Psychological safety face-to-face #5	*During face-to-face interactions, it is important to me that interactions in the Agile community are psychologically safe.*	Likert (1–5)	(as above)
Psychological safety face-to-face #6	*During face-to-face interactions, I experience psychological safety in the Agile community to be high.*	Likert (1–5)	(as above)

Table A.24: Phase I questionnaire: open-ended questions

Variable	Item (stem in italics)	Type
Experience with low psychological safety	*What are examples of behavior you have personally encountered during interactions in the Agile community that lowered psychological safety for you?*	Essay
Factors that decrease psychological safety	*In your experience, what behavior or factors decrease psychological safety in the Agile community?*	Essay
Effects of low psychological safety	*How does a lack of psychological safety in interactions in the Agile community affect you or your behavior?*	Essay
Strategies to improve psychological safety	*What measures or strategies could improve psychological safety and encourage more respectful and constructive discussions within the Agile community?*	Essay

Table A.25: Phase I questionnaire: demographic and background variables

Variable	**Item (stem in italics)**	**Type / response options**
Role	*What best describes your role?*	Categorical: Developer / analyst / tester / designer; Scrum Master; Agile Coach; Trainer or facilitator; Consultant; Other; Prefer not to say
Age group	*What is your age group?*	Categorical: 18-25; 26-35; 36-45; 46-55; 56-65; 66+; Prefer not to say
Seniority	*How many years have you been active in the Agile community?*	Categorical: Less than a year; 1–2 years; 2–5 years; 5–10 years; More than 10 years
Gender	*What is your gender?*	Categorical: Male; Female; Non-binary or third gender; Prefer not to say
Content creatorship	*Do you create content for the Agile community (videos, blogs, products, etc.)?*	Categorical: Yes; No

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Table A.26: Phase II Questionnaire

Variable	Question	Type	Options
Psychological safety face-to-face vs online	How do these results align with your expectations?	Likert (1-5)	The results strongly differ from my expectations, The results somewhat differ from my expectations, The results neither differ from nor match my expectations, The results somewhat match my expectations, The results strongly match my expectations
	What is your interpretation of this result?	Essay	n/a
Psychological safety by gender	How do these results align with your expectations?	Likert (1-5)	The results strongly differ from my expectations, The results somewhat differ from my expectations, The results neither differ from nor match my expectations, The results somewhat match my expectations, The results strongly match my expectations
	What is your interpretation of this result?	Essay	n/a
Psychological safety by role	How do these results align with your expectations?	Likert (1-5)	The results strongly differ from my expectations, The results somewhat differ from my expectations, The results neither differ from nor match my expectations, The results somewhat match my expectations, The results strongly match my expectations
	What is your interpretation of this result?	Essay	n/a
Psychological safety by age	How do these results align with your expectations?	Likert (1-5)	The results strongly differ from my expectations, The results somewhat differ from my expectations, The results neither differ from nor match my expectations, The results somewhat match my expectations, The results strongly match my expectations
	What is your interpretation of this result?	Essay	n/a
Follow-up questions	Based on the results you've interpreted so far, what follow-up questions come to mind for you?	Essay	n/a
Prior participation in Phase I	Did you participate in the survey on psychological safety yourself?	Categorical	Yes, No, I do not remember
Interest in research process	Did you find this process interesting?	Likert (1-5)	Not interesting at all, Slightly interesting, Moderately interesting, Very interesting, Extremely interesting

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